

Nonlinear Operator Composition for Robust Reconstruction of Latent Psychometric Representations under Structured Perturbations

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Abstract

Psychometric scoring procedures commonly aggregate or transform observed questionnaire responses into scale- or facet-level scores. Their transparency and reproducibility are valuable, but deterministic scoring does not itself provide a learned reconstruction mechanism when the observed response vector is perturbed. In real questionnaire settings, unintended response deviations may arise from insufficient attention, fatigue, distraction, response uncertainty, or item ambiguity, while other deviations may be deliberate; the present study models a controlled subset of these phenomena through semi-synthetic structured transformations. We study this problem as a supervised nonlinear functional approximation task in which a corrupted observation is used to reconstruct the facet representation defined from the corresponding original response pattern.

We evaluate a multilayer perceptron (MLP) Gateway on the IPIP-NEO-120 and IPIP-NEO-300 configurations, selected as complementary 120- and 300-item instantiations of a documented 30-facet psychometric framework. The Gateway represents the reconstruction mapping as a composition of finite-dimensional nonlinear transformations. Its configurations were developed through controlled model-selection procedures using corrupted-validation performance as the primary robustness criterion, while the corruption mechanism and severity remained external evaluation descriptors. An immutable Gold-defined facet representation is computed from the original uncorrupted responses and remains unchanged after perturbation. Seven structured perturbation families are evaluated, spanning item-, facet-, category-, collapse-, and randomized response transformations.

The IPIP-NEO-120 validation benchmark contains 61,556 clean and 338,444 corrupted observations. The IPIP-NEO-300 validation benchmark contains 21,808 clean and 338,444 corrupted observations. In frozen validation, the IPIP-120 Gateway reduces corrupted-input RMSE from 0.462274 to 0.347833, corresponding to a 24.756% relative reduction, and increases Pearson and Spearman correlation with Gold, while traditional scoring retains lower corrupted MAE. In the final IPIP-300 validation audit, the Gateway reduces corrupted-input RMSE from 0.240123 to 0.188775, corresponding to a 21.384% relative reduction, increases Pearson and Spearman correlation with Gold, and yields lower row-level RMSE than traditional scoring for 53.452% of evaluated corrupted observations. Traditional scoring nevertheless retains lower corrupted MAE.

The IPIP-NEO-120 program additionally includes an independent held-out test benchmark. The IPIP-NEO-300 validation program is complemented by a mechanistic and geometric audit examining latent displacement, effective rank, spectral structure, centered kernel

alignment, neighborhood preservation, output deformation, item-to-output and facet-to-output propagation, output-error correlation, and regression-based error associations. The findings are interpreted as evidence of robustness to the specified semi-synthetic perturbation protocol rather than as evidence of clinical validity, diagnostic utility, or replacement of established psychometric measurement.

Keywords: psychometric reconstruction, personality assessment, nonlinear function approximation, structured perturbations, response corruption, robust machine learning, latent psychometric representation, IPIP-NEO, reproducible research

Statement of Relevance

Questionnaire scores are commonly treated as fixed summaries of observed responses. This study addresses a computational, statistical, and mathematical problem that differs from conventional psychometric scoring: when questionnaire responses are subjected to structured perturbations varying in magnitude and granularity, can a learned mapping reconstruct the canonical reference psychometric representation defined by the established IPIP-NEO scoring framework from the participant’s original, unperturbed response pattern (**goldberg1999ipip; johnson2014ipipneo120; kajonius2019ipipneo**)?

The proposed framework employs instrument-specific multilayer perceptron (MLP) Gateway configurations for the IPIP-NEO-120 and IPIP-NEO-300, developed through controlled architecture and objective selection procedures. Rather than relying exclusively on the items conventionally assigned to each facet, the Gateway operates on the complete item-response vector and learns cross-item relationships within the observational space.

Across the IPIP-NEO-120 and IPIP-NEO-300 experiments, the Gateway demonstrates improved performance across multiple complementary evaluation dimensions under the investigated semi-synthetic perturbation protocol, including lower global RMSE, higher Pearson and Spearman profile correlations, and lower RMSE when stratified by corruption family, perturbation severity, and facet. The magnitude of the improvement varies as a function of perturbation structure and severity. The principal metric-specific trade-off observed is that conventional scoring retains lower mean absolute error (MAE), despite the Gateway achieving lower RMSE and stronger profile-level correspondence under the evaluated corrupted conditions.

The perturbation framework is semi-synthetic and empirically grounded: selected transformations use observed response patterns from real incomplete human participants as donor material, while the canonical reference representation remains fixed and is derived from the corresponding original, unperturbed response profile. This design permits controlled manipulation of response structure without reducing the evaluation protocol to arbitrarily generated response values.

Taken together, the results provide evidence of improved reconstruction agreement with the established reference representation across several complementary error and profile-level metrics within the evaluated conditions. They therefore provide a methodological basis for investigating potential incremental utility, validity, and eventual clinical applicability relative to conventional scoring procedures. They do not, however, establish clinical utility, diagnostic accuracy, or external validity beyond the instruments, populations, and perturbation protocols evaluated in this study.

1. Introduction

Personality assessment seeks to translate observed questionnaire responses into interpretable representations of individual differences. Within the Five-Factor Model (FFM), the IPIP-NEO

family provides a public-domain operationalization of five broad domains and thirty narrower facet constructs. The IPIP-NEO-120 was developed as a 120-item short form retaining the thirty-facet structure of the longer 300-item inventory, with four items per facet (**johnson2014ipipneo120**). The broader FFM framework and its facet-level organization have been extensively examined in personality psychology (**mccrae1992fivefactor**; **kajonius2019ipipneo**).

The conventional scoring procedure is intentionally deterministic and transparent. Responses to negatively keyed items are reverse-scored, items designated for each facet are aggregated, and the resulting facet values can be reproduced directly from the observed response vector. This property is valuable for interpretability, auditability, and methodological reproducibility. It also defines a canonical reference mapping from a complete response vector to a thirty-dimensional psychometric representation. In the present study, we use the term *canonical reference psychometric representation* to denote the representation obtained by applying this established scoring framework to the participant’s original, unperturbed response pattern. This representation is an operational reference target for the reconstruction task and is not interpreted as an independently observed psychological truth or as a clinical ground truth.

The computational problem considered here differs from conventional psychometric scoring in that the model is not asked merely to compute the deterministic score associated with an observed response vector. Instead, the observed vector may have undergone a structured alteration before inference, and the task is to reconstruct the canonical reference representation that would have been obtained from the corresponding original response pattern. Let $\mathbf{X}_{\text{clean}} \in \mathbb{R}^d$ denote an original response vector, with $d \in \{120, 300\}$, and let $\mathcal{C}_\phi$ denote a structured perturbation operator parameterized by its transformation mechanism and severity. The observed input is therefore

$$\tilde{\mathbf{X}} = \mathcal{C}_\phi(\mathbf{X}_{\text{clean}}), \quad (1)$$

whereas the canonical reference representation is computed independently from the unperturbed vector,

$$= g(\mathbf{X}_{\text{clean}}) \in \mathbb{R}^{30}. \quad (2)$$

A learned Gateway seeks to approximate this reference mapping from the perturbed observation,

$$\hat{\mathbf{Y}} = \mathcal{F}_\theta(\tilde{\mathbf{X}}), \quad (3)$$

with the objective of minimizing the discrepancy between $\hat{\mathbf{Y}}$ and over the specified distribution of perturbations.

This formulation creates a distinction between the *observation space*, the *reference representation space*, and the *perturbation process*. The observation space remains the original item-response vector, where responses are recorded on the instrument’s five-point response scale. The reference space is the thirty-dimensional facet representation defined by the established scoring procedure. The perturbation process modifies only the observed input presented to the reconstruction model; it does not redefine the reference target. This separation is central to the experimental design because it allows reconstruction accuracy to be evaluated against a fixed participant-specific reference rather than against a target recomputed from the corrupted observation.

The perturbation protocol is deliberately structured across multiple levels of granularity and severity. The evaluated families include item-level substitution, facet-level transplantation, category-level transformations, collapse operations, and randomized item changes. Importantly,

the protocol is not restricted to arbitrarily generated numerical noise. For selected transformation families, donor material is drawn from empirically observed response patterns of real incomplete participants, while the transformation itself is applied under controlled experimental rules. Incomplete participants do not provide reference targets; their observed responses serve only as donor material, whereas the canonical reference remains associated with the original complete participant profile. :contentReference[oaicite:2]index=2 This design motivates the description *semi-synthetic, empirically grounded perturbation protocol*: the perturbation mechanism is experimentally controlled, while part of its response content is drawn from observed human data.

The reconstruction problem can therefore be viewed as a finite-dimensional nonlinear function-approximation task. Classical approximation theory establishes broad approximation capabilities for multilayer feedforward networks on finite-dimensional mappings (**hornik1989universal**). Accordingly, the MLP Gateway is represented as a finite composition of parameterized transformations,

$$\mathcal{F}_\theta = \mathcal{T}_L \circ \mathcal{T}_{L-1} \circ \cdots \circ \mathcal{T}_1, \quad (4)$$

where the individual transformations combine learned affine mappings, normalization, and nonlinear activation functions as specified by the architecture.

The terminology of *nonlinear operator composition* is used here in this finite-dimensional sense. We do not claim a neural operator in the stricter sense used in the neural-operator literature, where the objective is to learn maps between function spaces and the formulation is typically infinite-dimensional or discretization-aware (**kovachki2023neuraloperator**). The present formulation instead emphasizes the mathematical structure of the Gateway as a composition of nonlinear transformations acting on finite-dimensional questionnaire vectors.

Two complementary instrument configurations are investigated. The IPIP-NEO-120 experiment uses a compact MLP Gateway mapping 120 item responses to the thirty-dimensional reference representation. The IPIP-NEO-300 experiment uses a structured-output MLP Gateway mapping 300 item responses to the same thirty-dimensional representation through five six-facet output heads. The resulting configurations were developed through controlled model selection procedures rather than by treating the architecture as arbitrary; for IPIP-300, architecture screening and objective screening were performed as separate experimental procedures using corrupted-validation RMSE as the selection criterion, while the independent test population remained outside model selection. :contentReference[oaicite:3]index=3

The central empirical question is therefore:

Can a learned nonlinear mapping reconstruct the canonical reference psychometric representation more accurately than conventional deterministic scoring when both methods are applied to the same structured-perturbed questionnaire response vector?

The evaluation is intentionally multidimensional. Reconstruction is assessed not only through pointwise error, but also through profile-level correspondence and stratified behavior across perturbation families, severity levels, and facets. Pearson and Spearman correlations provide complementary information about correspondence between the predicted and reference thirty-dimensional profiles, while RMSE and MAE characterize different aspects of numerical error. This distinction is important because a method may reduce large errors without simultaneously minimizing absolute error across all observations.

More broadly, robustness in machine learning is not a single property but depends on the class and magnitude of perturbation under consideration. Recent work has emphasized the need to characterize robustness with respect to specific perturbation and failure regimes

rather than treating performance under one perturbation distribution as universal evidence of robustness (**balendran2025robustness**). Our contribution is correspondingly bounded: we study robustness to a defined family of structured perturbations in questionnaire response space and examine whether a learned nonlinear mapping can preserve the canonical reference representation under those conditions.

The study is methodological and does not assume that successful numerical reconstruction establishes clinical validity, diagnostic accuracy, or general-purpose psychometric superiority. Instead, it investigates whether cross-item information learned from complete response vectors can provide a robust reconstruction mechanism when observations depart from their original configuration under controlled, empirically grounded perturbation processes.

2. Problem Formulation

2.1 Observation Space

Let

$$\mathbf{X}_{\text{clean}} = (x_1, \dots, x_d), \quad x_i \in \{1, 2, 3, 4, 5\}, \quad (5)$$

denote the complete item-response vector of a participant, where $d \in \{120, 300\}$ according to the instrument configuration. The corresponding observation space is therefore the finite discrete space

$$\mathcal{X}_d = \{1, 2, 3, 4, 5\}^d. \quad (6)$$

The same participant-level response vector is used to define both the unperturbed reference representation and the perturbed observations used for robustness evaluation.

2.2 Canonical Reference Psychometric Representation

Let

$$g : \mathcal{X}_d \rightarrow \mathbb{R}^{30} \quad (7)$$

denote the deterministic scoring transformation defined by the established IPIP-NEO scoring framework. The resulting representation is denoted by

$$= g(\mathbf{X}_{\text{clean}}) \in \mathbb{R}^{30}, \quad (8)$$

and is referred to throughout this manuscript as the *canonical reference psychometric representation*.

For a negatively keyed item, the conventional reverse-scoring transformation is

$$x_i^{\text{score}} = 6 - x_i. \quad (9)$$

For non-reversed items,

$$x_i^{\text{score}} = x_i. \quad (10)$$

Let $\mathcal{I}_j$ denote the set of item indices assigned to facet j , with $j \in \{1, \dots, 30\}$. The reference value for facet j is then

$$y_j = \frac{1}{|\mathcal{I}_j|} \sum_{i \in \mathcal{I}_j} x_i^{\text{score}}. \quad (11)$$

Consequently,

$$= (y_1, \dots, y_{30}). \quad (12)$$

For IPIP-NEO-120, each facet contains four designated items; for IPIP-NEO-300, each facet contains ten designated items. The scoring transformation is deterministic and is applied to the original, unperturbed response vector.

For notational purposes, the reference representation may equivalently be written as an affine transformation,

$$= A\mathbf{X}_{\text{clean}} + \mathbf{b}, \quad (13)$$

where $A \in \mathbb{R}^{30 \times d}$ encodes item-to-facet membership, aggregation weights, and reverse-key direction, while $\mathbf{b} \in \mathbb{R}^{30}$ contains the corresponding constants induced by reverse scoring.

This formulation makes explicit that the canonical reference representation is a fixed deterministic transformation of the original response vector. It is an operational reference for reconstruction and is not interpreted as an independently observed psychological truth or clinical ground truth.

2.3 Structured Semi-Synthetic Perturbation Process

A perturbation realization is represented by a parameter

$$\phi = (c, s, \omega), \quad (14)$$

where c denotes the perturbation family, s denotes its severity or magnitude, and ω contains the realization-specific structural information, including selected items, facets, categories, collapse configurations, and, where applicable, donor assignments.

The corresponding perturbation operator is

$$\mathcal{C}_\phi : \mathcal{X}_d \rightarrow \mathcal{X}_d, \quad (15)$$

and produces the observed perturbed response vector

$$\tilde{\mathbf{X}} = \mathcal{C}_\phi(\mathbf{X}_{\text{clean}}). \quad (16)$$

The perturbation protocol is semi-synthetic and empirically grounded. For donor-based transformations, response values are drawn from observed response patterns of real incomplete participants assigned to the corresponding data partition. These participants do not provide psychometric reference targets; their observed responses serve only as donor material for the construction of controlled perturbations.

Accordingly, the perturbation process modifies the observation while leaving the participant-specific reference representation unchanged:

$$\boxed{\mathbf{X}_{\text{clean}} \rightarrow} \quad \text{and} \quad \boxed{\tilde{\mathbf{X}} = \mathcal{C}_\phi(\mathbf{X}_{\text{clean}}) \rightarrow}. \quad (17)$$

The second mapping is the reconstruction problem studied in this work. The reference target is never recomputed from $\tilde{\mathbf{X}}$.

2.4 Separation of Predictive Inputs and Perturbation Metadata

The predictive input available to the learned Gateway is exclusively the perturbed response vector $\tilde{\mathbf{X}}$. Perturbation metadata are not supplied to the model.

Formally,

$$\text{Input}_{\text{Gateway}} = \tilde{\mathbf{X}}, \quad (18)$$

whereas

$$c, \quad s, \quad \omega \quad (19)$$

are retained externally for dataset construction, auditing, stratified evaluation, and error analysis.

Thus, the model cannot directly condition its prediction on the known perturbation family or severity label.

2.5 Functional Reconstruction

The learned Gateway defines a parametric nonlinear mapping

$$\mathcal{F}_\theta : \mathcal{X}_d \rightarrow \mathbb{R}^{30}, \quad (20)$$

such that

$$\hat{\mathbf{Y}} = \mathcal{F}_\theta(\tilde{\mathbf{X}}). \quad (21)$$

The reconstruction objective is therefore to approximate the canonical reference representation from the perturbed observation:

$$\hat{\mathbf{Y}} \approx . \quad (22)$$

More formally, given a distribution over complete response vectors and perturbation realizations, the population objective is

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{\mathbf{X}_{\text{clean}}, \phi} [\mathcal{L}(\mathcal{F}_\theta(\mathcal{C}_\phi(\mathbf{X}_{\text{clean}})), g(\mathbf{X}_{\text{clean}}))]. \quad (23)$$

The learned mapping is therefore optimized with respect to the participant-specific canonical reference associated with the unperturbed response vector rather than with respect to the perturbed observation itself.

2.6 Conventional Deterministic Comparator

The deterministic comparator applies the same established scoring function directly to the perturbed response vector:

$$= g(\tilde{\mathbf{X}}). \quad (24)$$

For clean observations,

$$\tilde{\mathbf{X}} = \mathbf{X}_{\text{clean}} \implies =, \quad (25)$$

by construction.

For perturbed observations, the comparison is therefore between

$$= g(\tilde{\mathbf{X}}) \quad (26)$$

and

$$\hat{\mathbf{Y}} = \mathcal{F}_\theta(\tilde{\mathbf{X}}), \quad (27)$$

with both evaluated against the same canonical reference

$$= g(\mathbf{X}_{\text{clean}}). \quad (28)$$

This establishes the head-to-head evaluation used throughout the study: both methods receive the same perturbed response vector, while the reference representation is fixed by the participant’s original unperturbed response pattern.

3. MLP Gateway Framework

3.1 Composed Nonlinear Mapping

The Gateway is formulated as a finite-dimensional parametric nonlinear mapping from the observed questionnaire response space to the canonical reference psychometric representation space. Let

$$\mathcal{N} : \mathcal{X}_d \rightarrow [-1, 1]^d \quad (29)$$

denote the element-wise input normalization

$$\mathcal{N}(x_i) = \frac{x_i - 3}{2}, \quad (30)$$

which maps the original five-point response scale $\{1, 2, 3, 4, 5\}$ to the interval $[-1, 1]$.

The Gateway can then be written as the composition

$$\mathcal{F}_\theta = \mathcal{G}_\theta \circ \mathcal{N}, \quad (31)$$

where $\mathcal{G}_\theta$ denotes the learned multilayer transformation. More generally, the nonlinear component is expressed as

$$\mathcal{G}_\theta = \mathcal{T}_L \circ \mathcal{T}_{L-1} \circ \cdots \circ \mathcal{T}_2 \circ \mathcal{T}_1. \quad (32)$$

For a hidden layer ℓ , the pre-activation state is

$$\mathbf{z}_\ell = W_\ell \mathbf{h}_{\ell-1} + \mathbf{b}_\ell, \quad (33)$$

followed, where specified, by normalization, nonlinear activation, and dropout:

$$\mathbf{h}_\ell = \text{Dropout}_\ell(\text{GELU}(\text{LayerNorm}_\ell(\mathbf{z}_\ell))). \quad (34)$$

The final output layer produces

$$\hat{\mathbf{Y}} = W_{\text{out}} \mathbf{h}_L + \mathbf{b}_{\text{out}}, \quad (35)$$

with no output activation or clipping imposed in the reported inference pipeline.

This construction represents the Gateway as a finite composition of parameterized affine maps and nonlinear transformations. The resulting mapping is therefore explicitly nonlinear while remaining defined on a finite-dimensional item-response space.

3.2 Instrument-Specific Gateway Configurations

The study uses two instrument-specific Gateway configurations. Both map the complete item-response vector to the same thirty-dimensional canonical reference representation, but they differ in input dimensionality and internal output organization.

The IPIP-NEO-120 configuration operates on 120 item responses, whereas the IPIP-NEO-300 configuration operates on 300 item responses. The shared 30-dimensional output space makes the two experiments directly comparable at the level of the reconstructed psychometric representation, while the instrument-specific architectures acknowledge the different observational dimensionalities of the two questionnaires.

3.3 IPIP-NEO-120 Gateway

The IPIP-NEO-120 Gateway uses the compact mapping

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 64 \rightarrow 30. \quad (36)$$

The three hidden transformations use Layer Normalization, GELU activations, and dropout with probability $p = 0.10$. The final layer maps the 64-dimensional hidden representation directly to the 30 facet outputs.

The complete configuration contains 58,910 trainable parameters.

The model therefore implements

$$\mathcal{F}_\theta^{(120)} : \mathcal{X}_{120} \rightarrow \mathbb{R}^{30}. \quad (37)$$

The architecture was evaluated under the same response-space corruption framework used for the subsequent robustness benchmark. The corresponding model-selection procedure treated corrupted-validation performance as the primary robustness criterion, while keeping the independent TEST population outside model selection.

3.4 IPIP-NEO-300 Gateway

The IPIP-NEO-300 Gateway uses a shared trunk followed by structured output heads:

$$300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow \{6, 6, 6, 6, 6\}. \quad (38)$$

The shared hidden trunk consists of three transformations with widths 256, 128, and 64. Each of the five output heads receives the same 64-dimensional shared representation and predicts six facet values, giving

$$5 \times 6 = 30 \quad (39)$$

output dimensions in total.

The resulting mapping is

$$\mathcal{F}_\theta^{(300)} : \mathcal{X}_{300} \rightarrow \mathbb{R}^{30}. \quad (40)$$

The configuration contains 121,054 trainable parameters.

The structured-output design preserves a shared representation across all facets while imposing an explicit partition of the final output layer into five six-facet groups. This organization corresponds to the five broad Five-Factor domains represented by the thirty-facet output structure.

3.5 Objective Function

The Gateway is optimized against the canonical reference representation using a composite objective that combines pointwise reconstruction error with profile-level correspondence:

$$\mathcal{L}_H = 0.90\mathcal{L}_{\text{MSE}} + 0.10\mathcal{L}_{\text{Pearson}}. \quad (41)$$

For a thirty-dimensional reference vector,

$$\mathcal{L}_{\text{MSE}} = \frac{1}{30} \sum_{j=1}^{30} \left(\hat{y}_j - y_j^{\text{ref}} \right)^2. \quad (42)$$

The profile-correlation component is defined as

$$\mathcal{L}_{\text{Pearson}} = 1 - r_{\text{Pearson}}, \quad (43)$$

where r_{Pearson} denotes the Pearson correlation between the predicted and canonical reference thirty-dimensional profiles.

For numerical stability, the correlation computation uses a stabilizing constant

$$\varepsilon = 10^{-8}. \quad (44)$$

The objective therefore jointly penalizes deviations in numerical magnitude while encouraging preservation of the multivariate profile structure. Pearson and Spearman correlations are not used interchangeably: Pearson contributes to the training objective, whereas Pearson and Spearman are both retained as evaluation metrics.

3.6 Subject-Isolated Exposure Schedule

Training uses a deterministic Subject-Isolated Exposure Schedule (SIES). The schedule assigns each complete training participant 143 views, consisting of

$$3 \text{ clean} + 7 \times 20 \text{ corrupted} = 143 \quad (45)$$

views per participant.

The value of 143 should be understood as an operational design choice rather than as a theoretically optimal number of exposures. The number of views per participant was constrained by the available computational resources during dataset construction, particularly memory capacity, as well as by the subsequent training time required to process the resulting dataset. The schedule was therefore selected as a reproducible compromise that preserved representation of all seven perturbation families, multiple corruption realizations per family, and repeated clean observations without making the training corpus and optimization cycle unnecessarily expensive.

The seven corruption families are represented with twenty variants per family for each participant. The resulting exposure structure is

$$\left\{ \begin{array}{l} \text{CLEAN,} \\ \text{ITEM_SWAP,} \\ \text{FACET_SWAP,} \\ \text{CATEGORY_PARTIAL_SWAP,} \\ \text{CATEGORY_SWAP,} \\ \text{ITEM_COLLAPSE,} \\ \text{CATEGORY_COLLAPSE,} \\ \text{RANDOM_ITEM_CHANGE} \end{array} \right\}. \quad (46)$$

The three clean views serve as repeated clean anchors within the participant block, whereas the twenty realizations per corruption family provide repeated exposure to distinct perturbation configurations. These counts were not treated as estimates of an intrinsic optimal sampling ratio; rather, they define the fixed exposure schedule used for the reported training runs.

Within an optimization batch, duplicate views originating from the same underlying participant are prohibited. This constraint is intended to prevent multiple highly correlated views of the same participant from simultaneously occupying a training batch and to preserve subject-level separation within each exposure cycle.

The corruption family, severity, donor identity, and related construction metadata remain external to the predictive input space. The Gateway receives the response vector alone.

3.7 Architecture and Objective Selection

The Gateway configurations were not specified solely as arbitrary multilayer perceptrons. Model development used controlled selection procedures in which architecture and optimization objective were treated as experimentally distinct factors.

For IPIP-300, architecture screening evaluated seven candidate architectures on a 10,000-subject screening subset using corrupted-validation RMSE as the selection criterion. The structured-output architecture was selected from this procedure. A separate objective-screening experiment evaluated nine training objectives under a fixed compact architecture and identified the 90% MSE + 10% Pearson objective as the objective-screening champion. The independent TEST population remained reserved and was not used for either screening procedure.

This separation allows the architectural and objective contributions to be described independently rather than attributing all observed differences to a single undifferentiated model change.

The selected IPIP-300 Gateway uses a structured-output architecture with a 300-dimensional input representation followed by hidden representations of 256, 128, and 64 dimensions and five parallel output heads of six facets each. The first hidden transformation is

$$\mathbf{h}^{(256)} = \phi \left(\mathbf{W}^{(1)} \mathbf{x} + \mathbf{b}^{(1)} \right), \quad (47)$$

where

$$\mathbf{W}^{(1)} \in \mathbb{R}^{256 \times 300}, \quad \mathbf{b}^{(1)} \in \mathbb{R}^{256}. \quad (48)$$

The vector $\mathbf{b}^{(1)}$ therefore contains one learned affine offset for each neuron of the first hidden representation. Its parameter-space distribution is reported as an architectural diagnostic of the selected Gateway.

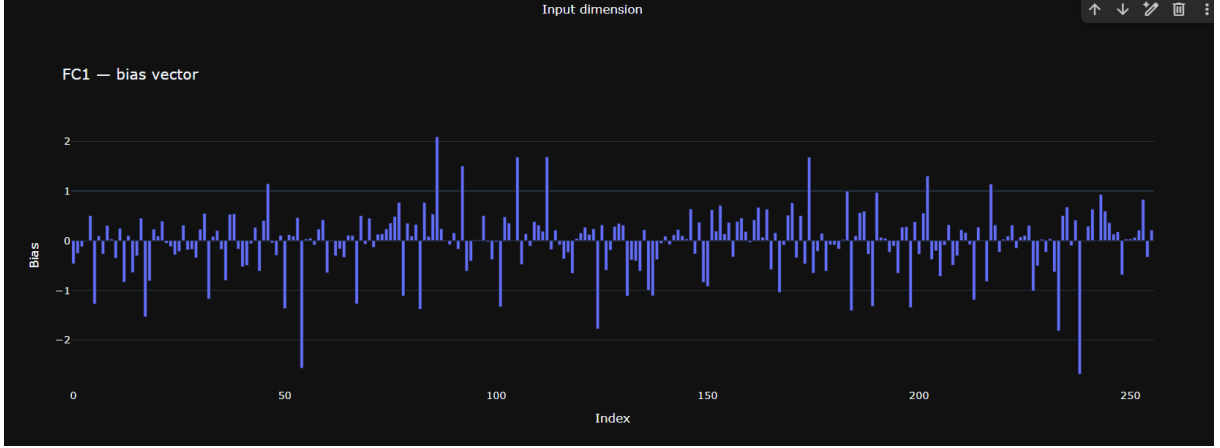


Figure 1: Learned bias vector of the first fully connected layer (FC1) of the IPIP-NEO-300 MLP Gateway. Each bar corresponds to one of the 256 neurons in the first hidden representation. The distribution represents the learned affine offsets of the selected production architecture.

For neuron j in the first hidden layer, the pre-activation value is

$$z_j = \sum_{i=1}^{300} W_{ji}^{(1)} x_i + b_j^{(1)}, \quad j = 1, \dots, 256. \quad (49)$$

The bias distribution is interpreted solely as a parameter-level diagnostic. It is not treated as a direct measure of neuron importance, item importance, or psychological meaning.

The learned parameters of the selected Gateway can be complemented by examining the numerical behavior of its hidden representations under inference. The three hidden stages have dimensions 256, 128, and 64, respectively, and their activation distributions provide a direct description of the internal numerical state induced by the evaluated response vectors.

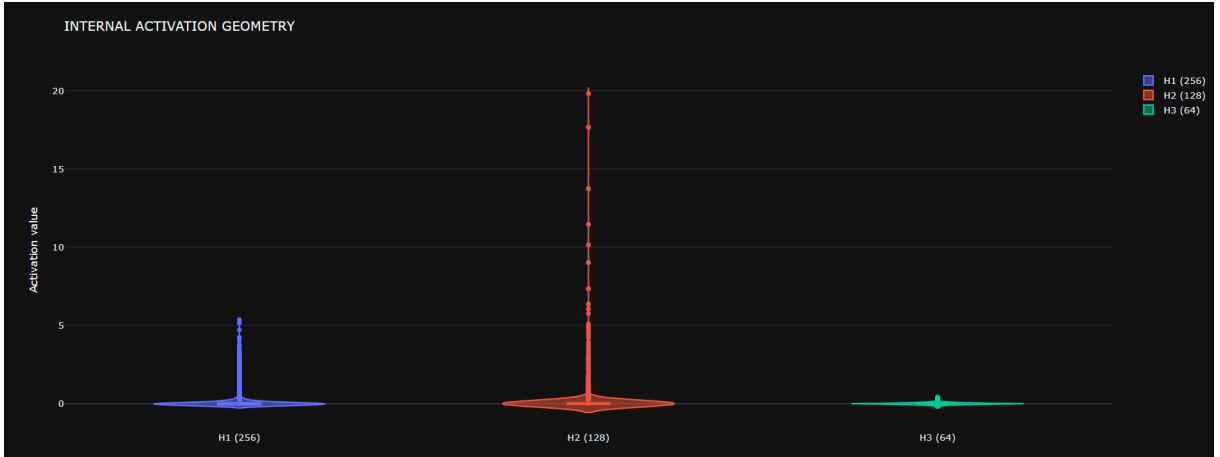


Figure 2: Internal activation distributions across the three hidden representations of the IPIP-NEO-300 MLP Gateway. H1, H2, and H3 correspond to the 256-, 128-, and 64-dimensional hidden stages, respectively.

For a representative clean observation, the 30-dimensional Gateway output is compared directly with its canonical reference profile. Let

$$\mathbf{Y}_{Gateway}^{clean} = \mathcal{F}_{\theta}(\mathbf{X}_{clean}), \quad (50)$$

and let $\mathbf{Y}^{ref}$ denote the corresponding canonical reference representation. The profile comparison provides a direct coordinate-level view of the agreement between the learned reconstruction and the reference representation.

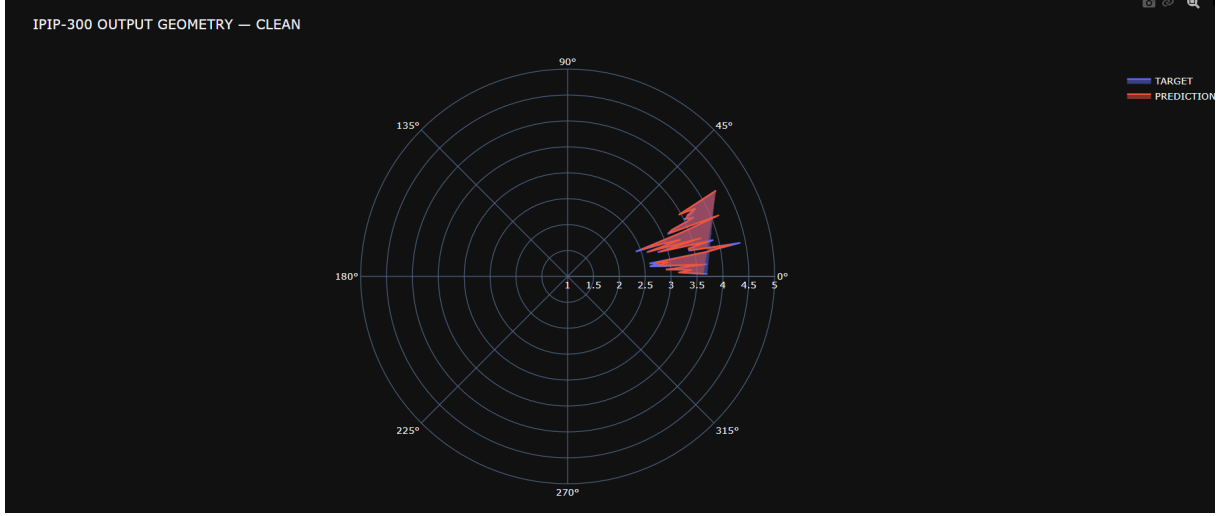


Figure 3: IPIP-NEO-300 output geometry for a representative clean observation. The Gateway prediction is shown together with the canonical reference across the 30-dimensional psychometric representation.

The same representation is evaluated after structured corruption of the observed response vector. For the corresponding corrupted observation $\tilde{\mathbf{X}}$, the Gateway produces

$$\mathbf{Y}_{Gateway}^{corr} = \mathcal{F}_\theta(\tilde{\mathbf{X}}), \quad (51)$$

while the canonical reference remains fixed at

$$\mathbf{Y}^{ref} = g(\mathbf{X}_{clean}). \quad (52)$$

This permits the clean and corrupted output profiles to be compared under the same reference construction without recomputing the target from the perturbed response.

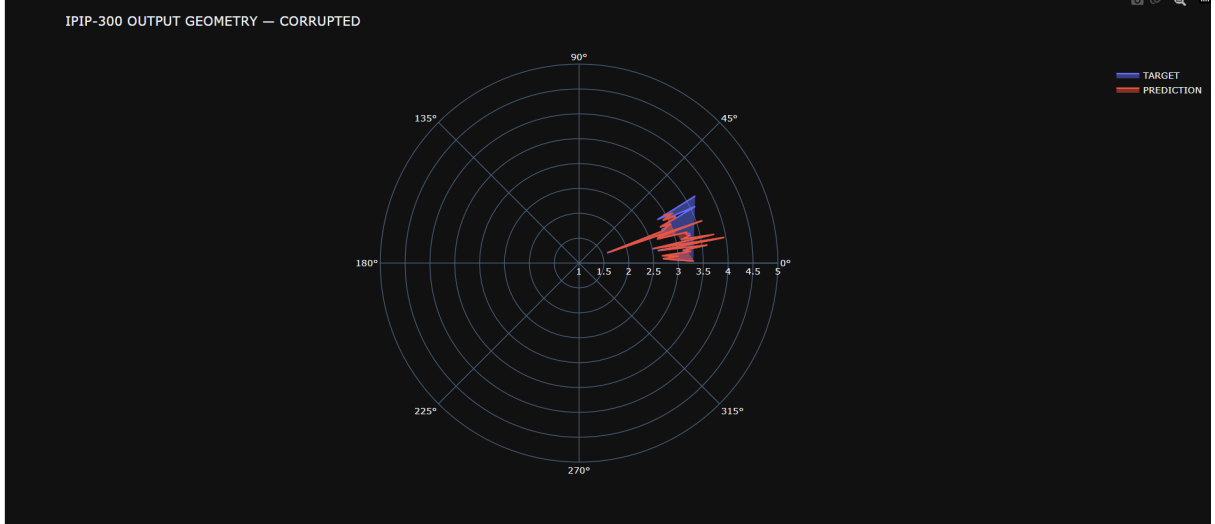


Figure 4: IPIP-NEO-300 output geometry for a representative corrupted observation. The Gateway prediction is shown relative to the unchanged canonical reference across the 30-dimensional psychometric representation.

The clean and corrupted profiles therefore provide complementary descriptive views of output-space behavior. They do not constitute additional performance benchmarks; quantitative reconstruction performance is evaluated separately using the predefined error and correlation metrics.

The selected IPIP-300 Gateway uses a structured-output architecture with a 300-dimensional input representation followed by hidden representations of 256, 128, and 64 dimensions and five parallel output heads of six facets each. The first hidden transformation is

$$\mathbf{h}^{(256)} = \phi \left(\mathbf{W}^{(1)} \mathbf{x} + \mathbf{b}^{(1)} \right), \quad (53)$$

where

$$\mathbf{W}^{(1)} \in \mathbb{R}^{256 \times 300}, \quad \mathbf{b}^{(1)} \in \mathbb{R}^{256}. \quad (54)$$

The subsequent hidden transformations are

$$\mathbf{h}^{(128)} = \phi \left(\mathbf{W}^{(2)} \mathbf{h}^{(256)} + \mathbf{b}^{(2)} \right), \quad (55)$$

$$\mathbf{h}^{(64)} = \phi \left(\mathbf{W}^{(3)} \mathbf{h}^{(128)} + \mathbf{b}^{(3)} \right), \quad (56)$$

followed by five parallel output heads,

$$\hat{\mathbf{Y}} = [\mathbf{H}_1, \mathbf{H}_2, \mathbf{H}_3, \mathbf{H}_4, \mathbf{H}_5], \quad (57)$$

with

$$\mathbf{H}_k \in \mathbb{R}^6, \quad k = 1, \dots, 5. \quad (58)$$

The resulting production network contains 778 neurons across the input, hidden, and output layers and 119,680 weighted inter-neuron connections, with 121,054 trainable parameters including affine biases and normalization parameters.

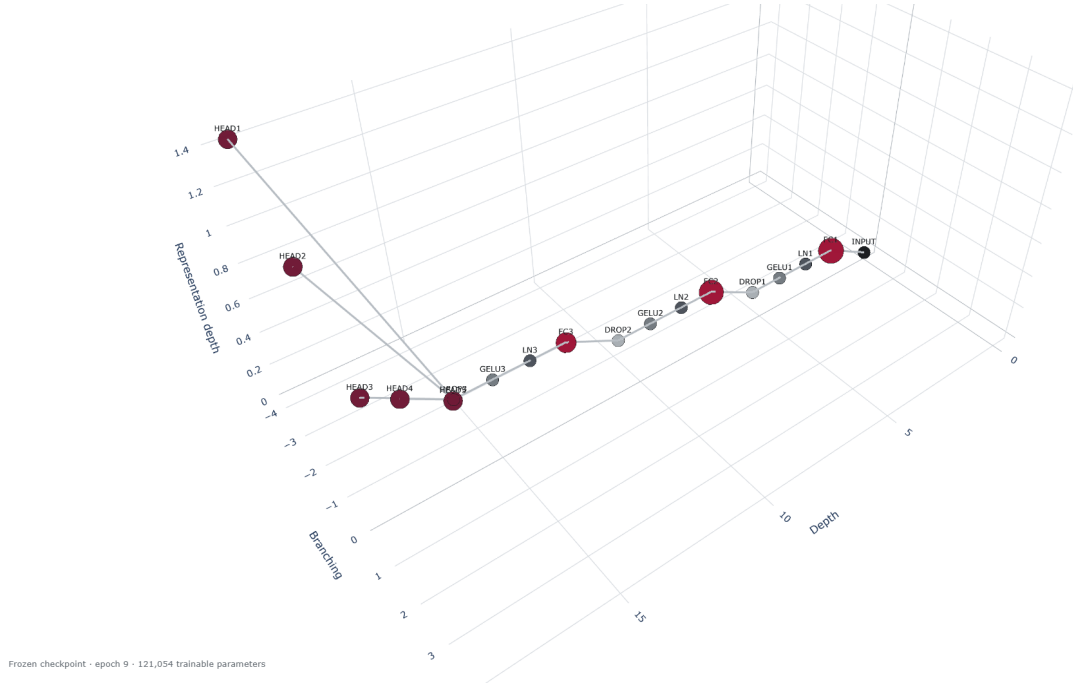
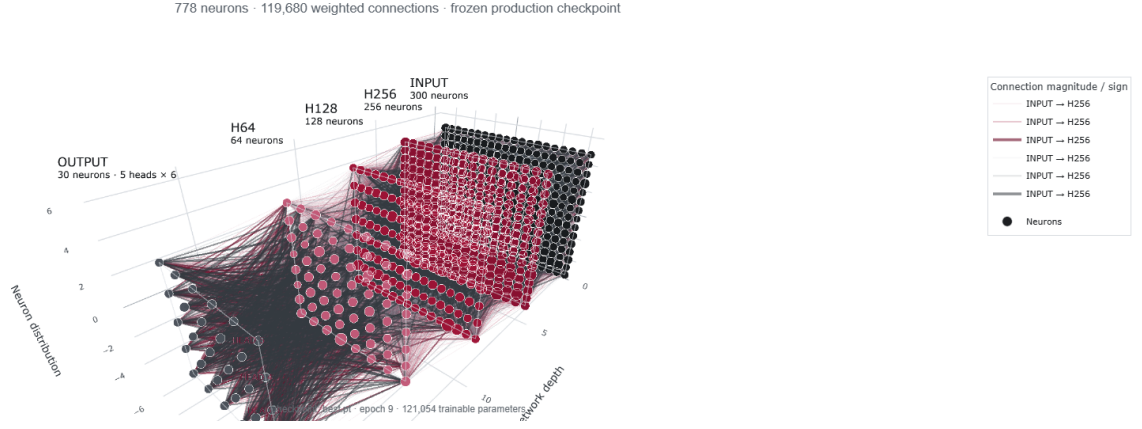
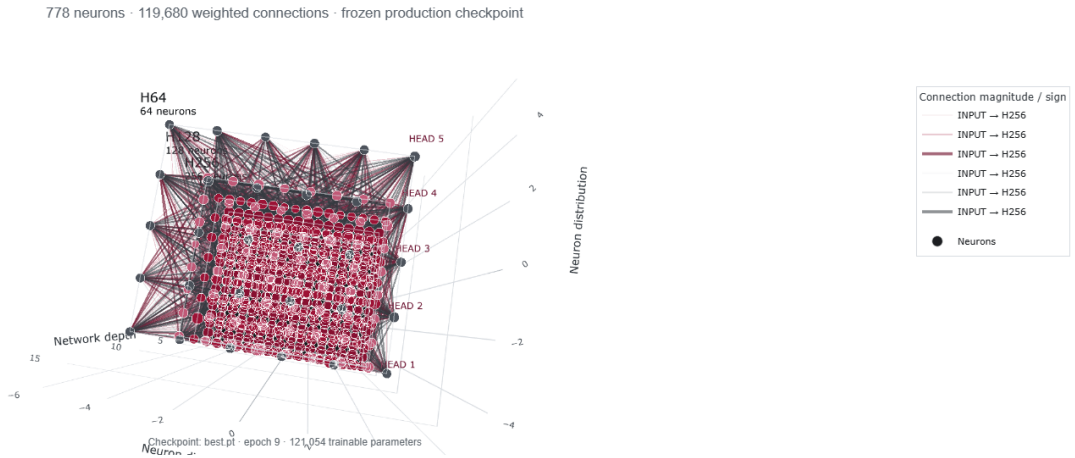


Figure 5: Three-dimensional graph representation of the frozen IPIP-NEO-300 MLP Gateway checkpoint. The graph depicts the complete neuron-level architecture from the 300-dimensional input through the 256-, 128-, and 64-dimensional hidden representations to the five six-dimensional output heads. The visualization represents 778 neurons and 119,680 weighted connections in the selected production checkpoint.

The graph provides an explicit structural representation of the selected network and complements the algebraic specification above. It is a visualization of the frozen computational architecture rather than an additional performance evaluation. The architectural graph can be examined at a finer level by representing the individual neurons and weighted connections of the frozen production checkpoint. This view preserves the complete layer structure while exposing the connectivity pattern between successive computational stages.



(a) Primary three-dimensional view of the frozen production checkpoint.



(b) Complementary three-dimensional viewing angle of the same frozen production checkpoint.

Figure 6: Neuron-level graph representation of the frozen IPIP-NEO-300 MLP Gateway checkpoint. The two panels show the same computational graph from complementary viewing angles, comprising 778 neurons and 119,680 weighted inter-neuron connections across the input, hidden, and structured output layers. The visualizations expose the full connectivity structure of the selected production checkpoint rather than a reduced schematic representation.

The two views represent the same frozen computational graph and therefore do not constitute independent observations or additional model evaluations. They are presented at near-full page width to preserve the visibility of layer structure, neuron organization, connection density, and output-head branching.

3.8 Inference Contract and Information Isolation

At inference time, the Gateway receives only the perturbed item-response vector:

$$\text{Gateway input} = \tilde{\mathbf{X}}. \quad (59)$$

No participant identifier, donor identifier, corruption-family label, severity code, perturbation pattern identifier, or reference representation is supplied to the model.

The information flow can therefore be summarized as

$$\boxed{\tilde{\mathbf{X}} \xrightarrow{\mathcal{N}} [-1, 1]^d \xrightarrow{\mathcal{G}_\theta} \mathbb{R}^{30} \rightarrow \hat{\mathbf{Y}}}. \quad (60)$$

The canonical reference representation enters only as a training target and as an evaluation reference. It is never provided as an inference input.

3.9 Output Range and Non-Clipping Policy

The Gateway does not apply post-hoc clipping of its outputs to the nominal questionnaire scoring interval. Consequently, predicted values outside the nominal $[1, 5]$ range remain observable and are retained as part of the model diagnostic analysis.

This policy prevents output clipping from concealing the learned model’s native error distribution and allows out-of-range predictions to be quantified separately from the primary reconstruction metrics.

4. Experimental Design

4.1 Instruments and Experimental Scope

The present study evaluates two complementary configurations of the IPIP-NEO family: the 300-item public-domain IPIP-NEO inventory and the 120-item short form developed by Johnson. The IPIP-NEO-120 was explicitly developed as a shorter form of the 300-item IPIP-NEO while retaining its thirty-facet Five-Factor Model organization, with four items per facet (**goldberg1999ipip; johnson2014ipipneo120**). The full IPIP-NEO configuration contains ten items per facet across the same thirty-facet organization (**goldberg1999ipip**).

The two configurations are treated as complementary instances of the same reconstruction problem rather than as identical architectural replications. Both preserve a thirty-dimensional reference representation, while differing in the dimensionality and redundancy of the observed item-response space. Accordingly,

$$d_{120} = 120, \quad d_{300} = 300, \quad p = 30, \quad (61)$$

where d denotes the input dimensionality and p denotes the dimensionality of the canonical reference representation.

The present empirical evaluation is restricted to the IPIP-NEO-120 and IPIP-NEO-300 configurations. The Gateway framework is designed as an extensible methodology for other established multidimensional personality inventories. Planned extensions include the HEXACO Personality Inventory–Revised (HEXACO-PI-R), particularly its 100-item and 200-item configurations, as well as other psychometrically documented instruments with established scoring frameworks (**lee2018hexaco; ashton2008hexaco**).

These extensions constitute future work and are not used to support any empirical claim reported in the present manuscript.

Table 1: Instrument and Gateway configurations evaluated in the present study.

Configuration	Reference basis	Items	Domains	Facets	Items/Facet	Gateway Parameters
IPIP-120	Johnson (2014) short form	120	5	30	4	58,910
IPIP-300	Goldberg (1999) IPIP-NEO	300	5	30	10	121,054

The two configurations therefore differ in observational dimensionality while sharing the same thirty-facet output structure. They should be interpreted as complementary evaluations of the reconstruction formulation, not as literal replications of a single architecture (johnson2014ipipneo120).

For both configurations, the deterministic comparator is the conventional instrument scoring procedure defined by the corresponding item keys and facet-item assignments. The Gateway receives the complete observed item-response vector and maps it to the same thirty-dimensional canonical reference space.

The exact architecture, parameter count, preprocessing procedure, random seeds, optimization settings, and model version are recorded in the reproducibility code and associated model metadata. The reported parameter counts therefore identify the evaluated model configurations rather than merely their nominal layer widths.

4.2 Traditional Scoring Baseline

Traditional scoring applies the established deterministic scoring rules of the corresponding instrument, including reverse scoring for negatively keyed items followed by aggregation within the designated facet-item sets. The same deterministic scoring implementation used to construct the canonical reference psychometric representation is used as the comparator.

For a clean observation,

$$\tilde{\mathbf{X}} = \mathbf{X}_{\text{clean}} \implies = g(\tilde{\mathbf{X}}) = g(\mathbf{X}_{\text{clean}}) = . \quad (62)$$

Thus, exact agreement between traditional scoring and the canonical reference on clean observations is a property of the experimental construction rather than an empirical performance result.

For perturbed observations, the head-to-head comparison is instead defined by

$$= g(\tilde{\mathbf{X}}), \quad \hat{\mathbf{Y}} = \mathcal{F}_\theta(\tilde{\mathbf{X}}), \quad (63)$$

with both outputs evaluated against the same canonical reference

$$= g(\mathbf{X}_{\text{clean}}). \quad (64)$$

This establishes a direct comparison in which both methods receive the same observed input while differing in the transformation applied to that input.

4.3 Immutable Reference Construction

For every complete participant, the canonical reference psychometric representation is computed once from the original, unperturbed response vector:

$$= g(\mathbf{X}_{\text{clean}}). \quad (65)$$

The perturbation operator is subsequently applied only to the observed response vector:

$$\tilde{\mathbf{X}} = \mathcal{C}_\phi(\mathbf{X}_{\text{clean}}). \quad (66)$$

The reference representation is invariant under the subsequent perturbation:

$$\boxed{= g(\mathbf{X}_{\text{clean}}) \text{ and remains fixed for all } \tilde{\mathbf{X}} = \mathcal{C}_\phi(\mathbf{X}_{\text{clean}})}. \quad (67)$$

Consequently, the perturbation process cannot redefine, overwrite, or otherwise alter the participant-specific reference target. The target is neither recomputed from the perturbed vector nor modified according to the perturbation family, severity, donor identity, or realization-specific transformation.

The experimental information flow is therefore

$$\boxed{\mathbf{X}_{\text{clean}} \xrightarrow{g}} \quad \text{and} \quad \boxed{\mathbf{X}_{\text{clean}} \xrightarrow{c_\phi} \tilde{\mathbf{X}}}. \quad (68)$$

The two reconstruction procedures then operate on the identical perturbed observation:

$$\boxed{\tilde{\mathbf{X}} \rightarrow \begin{cases} = g(\tilde{\mathbf{X}}), \\ \hat{\mathbf{Y}} = \mathcal{F}_\theta(\tilde{\mathbf{X}}) \end{cases} \rightarrow}. \quad (69)$$

This invariant is central to the experimental design. It ensures that the task being evaluated is reconstruction toward a participant-specific reference established before perturbation, rather than rescoreing the perturbed response vector and treating the perturbed state as a new target.

4.4 Corruption Families

Seven perturbation families are evaluated:

1. ITEM_SWAP
2. FACET_SWAP
3. CATEGORY_PARTIAL_SWAP
4. CATEGORY_SWAP
5. ITEM_COLLAPSE
6. CATEGORY_COLLAPSE
7. RANDOM_ITEM_CHANGE

Each family is evaluated across a predefined severity structure. The resulting evaluation therefore distinguishes perturbation mechanism from perturbation magnitude and permits family-level and severity-level analysis.

4.5 IPIP-120 Validation and Independent Test

The IPIP-120 validation benchmark contains 61,556 clean observations and 338,444 corrupted observations.

The final independent IPIP-120 TEST benchmark contains 61,557 clean observations and 1,723,596 corrupted observations, for a total of 1,785,153 observations. The TEST population remained fully outside model training, hyperparameter selection, architecture selection, objective selection, and other model-development procedures.

4.6 IPIP-300 Validation

The IPIP-300 validation benchmark contains 21,808 clean observations and 338,444 corrupted observations, for a total of 360,252 observations.

An independent IPIP-300 TEST population was reserved during dataset construction but was not generated and evaluated as part of the final publication protocol because of the additional computational and time requirements associated with producing and analysing the benchmark. The final protocol instead combines the independent IPIP-120 TEST benchmark with the expanded IPIP-300 validation program and its mechanistic and geometric analyses.

This decision is treated as a limitation of the present study and does not imply that an independent IPIP-300 TEST benchmark would be unnecessary for a subsequent full-scale evaluation.

4.7 Evaluation Metrics

The primary error metric is root mean squared error:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i^{\text{ref}})^2}. \quad (70)$$

The secondary absolute error metric is mean absolute error:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i^{\text{ref}}|. \quad (71)$$

Profile-level correspondence is evaluated using Pearson and Spearman correlation. Pearson correlation is used both as an evaluation metric and as a component of the composite training objective, whereas Spearman correlation is retained exclusively as an evaluation measure of rank-order correspondence.

4.8 Reproducibility Controls

The final IPIP-300 Gateway was trained under the following core configuration:

Table 2: Core training and reproducibility configuration for the final IPIP-300 Gateway.

Parameter	Value
Optimizer	AdamW
Batch size	8,192
Learning rate	0.002
Weight decay	0.0001
Dropout	0.10
Gradient clipping	1.0
Mixed precision	BF16
MSE weight	0.90
Pearson weight	0.10
Global seed	20,260,920
Dataset seed	20,260,920
Loss/SIES seed	20,261,697
SIES rounds	143

The model checkpoint is frozen and identified by its cryptographic SHA-256 digest. The final IPIP-300 Gateway corresponds to architecture G (Structured Output) combined with objective H ($0.90 \mathcal{L}_{\text{MSE}} + 0.10 \mathcal{L}_{\text{Pearson}}$).

All seeds, architectural specifications, preprocessing rules, optimization parameters, checkpoint metadata, and integrity hashes are retained in the reproducibility package.

5. Results

5.1 IPIP-120 Validation

The IPIP-NEO-120 robustness evaluation was conducted on a frozen validation benchmark comprising 61,556 clean observations and 338,444 structured-perturbed observations, for a total of 400,000 evaluated observations. The benchmark contains seven perturbation families and thirteen evaluated severity values. Because the evaluated checkpoint was selected during model development using corrupted-validation RMSE, this analysis is reported as a *non-confirmatory validation robustness benchmark* rather than as an independent generalization test.

The evaluated Gateway checkpoint corresponds to the compact architecture

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 64 \rightarrow 30, \quad (72)$$

with 58,910 trainable parameters and the composite objective

$$\mathcal{L}_H = 0.90 \mathcal{L}_{\text{MSE}} + 0.10 \mathcal{L}_{\text{Pearson}}. \quad (73)$$

The checkpoint was evaluated at epoch 9 using CUDA and float32 inference. No output clipping was applied. The architecture, parameter count, and checkpoint configuration were independently verified before evaluation.

On clean observations, conventional deterministic scoring reproduced the canonical reference psychometric representation exactly, with

$$\text{RMSE} = 0 \quad (74)$$

and maximum absolute deviation equal to zero by numerical audit. This agreement is a construction property of the reference-generating scoring procedure and is therefore not treated as an empirical advantage in the robustness comparison.

The global head-to-head validation results are shown in Table ??.

Table 3: IPIP-NEO-120 frozen validation performance on clean observations.

Method	RMSE	MAE	Pearson	Spearman
Traditional	0.000000	0.000000	1.000000	1.000000
MLP Gateway	0.136102	0.092990	0.986004	0.985641

Table 4: IPIP-NEO-120 frozen validation performance on corrupted observations.

Method	RMSE	MAE	Pearson	Spearman
Traditional	0.462274	0.181624	0.818282	0.816762
MLP Gateway	0.347833	0.210576	0.891710	0.875251

Under corrupted inputs, the MLP Gateway reduced RMSE from 0.462274 under conventional scoring to 0.347833, corresponding to an absolute reduction of 0.114441 and a relative reduction of 24.756%. The corresponding profile-level correlations also increased: Pearson correlation rose from 0.818282 to 0.891710, while Spearman correlation increased from 0.816762 to 0.875251. The corresponding absolute increases in Pearson and Spearman correlation were 0.073428 and 0.058489, respectively.

The improvement was not uniform across all error criteria. Corrupted MAE increased from 0.181624 under conventional scoring to 0.210576 under the MLP Gateway, corresponding to an absolute increase of 0.028952, or approximately 15.941% relative to the conventional scorer. The observed robustness effect should therefore be interpreted as metric-specific rather than as a uniform reduction in every measure of prediction error.

The Gateway does not reproduce the canonical reference psychometric representation exactly on clean validation observations. Its clean RMSE was 0.136102 and its clean MAE was 0.092990, while its clean Pearson and Spearman correlations remained high at 0.986004 and 0.985641, respectively. This distinction is important because the reported robustness improvement under corruption cannot be attributed to an exact replication of the deterministic scoring function on the clean domain. Instead, the learned Gateway represents an approximate nonlinear mapping whose observed robustness advantage emerges under the perturbed conditions evaluated in this benchmark.

5.2 Corruption-Family Analysis

The robustness pattern was not confined to the aggregate validation metric. The MLP Gateway achieved lower RMSE than conventional scoring in all seven evaluated perturbation families. Relative RMSE reductions ranged from 7.091% for CATEGORY_PARTIAL_SWAP to 44.793% for CATEGORY_COLLAPSE.

The family-specific results are shown in Table 5.

Table 5: IPIP-NEO-120 corrupted-validation RMSE by perturbation family.

Perturbation family	Traditional RMSE	MLP Gateway RMSE	Relative RMSE reduction
ITEM_SWAP	0.311312	0.288860	7.212%
FACET_SWAP	0.355175	0.313356	11.774%
CATEGORY_PARTIAL_SWAP	0.283323	0.263232	7.091%
CATEGORY_SWAP	0.637445	0.517382	18.835%
ITEM_COLLAPSE	0.338333	0.272785	19.374%
CATEGORY_COLLAPSE	0.733339	0.404853	44.793%
RANDOM_ITEM_CHANGE	0.365996	0.300759	17.824%

Thus, within the defined perturbation protocol, the aggregate RMSE reduction was observed across all seven corruption families rather than being confined to a single perturbation mechanism. The magnitude of the reduction varied substantially across families, indicating that the response to corruption depended on the structure of the applied transformation.

5.3 Severity-Stratified Analysis

The Gateway also achieved lower RMSE at every evaluated severity value. Across the thirteen reported severity levels, relative RMSE reductions ranged from 4.825% to 30.703%.

The smallest relative improvement occurred at severity 6 (4.825%), whereas the largest occurred at severity 5 (30.703%). The corresponding reductions at severity values 1, 2, 3, 4, 10, 12, 15, 24, 36, 48, and 60 were also positive.

This result indicates that the aggregate robustness effect was not produced by a single extreme corruption regime. Lower Gateway RMSE was observed across the complete set of reported severity values used in the IPIP-120 validation protocol. At the same time, the variation in relative reduction across severity levels indicates that robustness was not constant as perturbation severity changed.

5.4 Facet-Level Analysis

The aggregate robustness effect was also distributed across the thirty output facets. The MLP Gateway achieved lower RMSE than conventional scoring for all thirty facets evaluated under the corrupted-validation benchmark.

Relative facet-level RMSE reductions ranged from 10.404% to 43.819%. The largest reduction was observed for C1_SelfEfficacy (43.819%), whereas the smallest was observed for O6_Liberalism (10.404%). The complete facet-level results are retained in the machine-readable benchmark output.

The observation that all thirty evaluated facets exhibited lower RMSE should be distinguished from equality of effect magnitude: the relative reductions varied across facets, indicating heterogeneous reconstruction performance within the output representation.

5.5 Row-Level Paired Analysis

A row-level paired comparison was conducted across all 338,444 corrupted validation observations. The MLP Gateway produced lower row-wise RMSE than conventional scoring in 57.135% of evaluated rows, whereas conventional scoring produced lower RMSE in 42.865%; no exact ties were observed.

The mean row-level RMSE difference, defined as

$$\Delta_i = RMSE_{i, \text{Gateway}} - RMSE_{i, \text{Traditional}}, \quad (75)$$

was -0.060243 . A two-sided Wilcoxon signed-rank test applied to the row-level paired differences yielded $p < 10^{-300}$, and the corresponding paired Cohen's d was -0.363 .

These inferential quantities are reported as row-level supplementary statistics rather than participant-level population inference because participant identifiers were not available in the evaluated validation export. Accordingly, they should not be interpreted as evidence based on independent participant-level observations, nor should the reported row-level significance be treated as a substitute for subject-level inference.

5.6 Output-Range Diagnostic

The Gateway was evaluated without post-hoc clipping of its predictions to the nominal $[1, 5]$ scoring interval. Under corrupted validation, the fraction of output elements falling outside this interval was 0.00101740 (approximately 0.102%). This diagnostic is reported explicitly rather than being concealed through output clipping.

The presence of a small fraction of out-of-range predictions indicates that the Gateway was evaluated in its unconstrained numerical output state. No post-processing operation was introduced that could otherwise alter the reported RMSE, MAE, Pearson, or Spearman values.

5.7 IPIP-120 Independent Test

An independent IPIP-NEO-120 TEST benchmark was generated after completion of the model-development protocol. The TEST population comprises 61,557 clean observations and 1,723,596 corrupted observations, for a total of 1,785,153 evaluated observations. The benchmark additionally contains 31,316 TEST donor subjects and 28 family-level corruption blocks.

The TEST evaluation uses the frozen Gateway checkpoint without retraining, hyperparameter tuning, architectural modification, or post-hoc adaptation to the TEST observations. The TEST benchmark is therefore treated separately from the validation benchmark used during model development.

The final global, corruption-family, family-by-severity, severity, and facet-level TEST metrics are to be reported directly from the frozen machine-readable TEST result record. No TEST performance value is inferred from the validation benchmark, and no value is reported here until the final TEST output has been independently captured and verified.

Table 6: IPIP-NEO-120 independent TEST benchmark. Exact metrics are reported from the frozen machine-readable TEST result record and are not inferred from the validation benchmark.

Method	Corrupted RMSE	Corrupted MAE	Pearson	Spearman
Traditional	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
MLP Gateway	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>

Table 7: IPIP-NEO-120 independent TEST benchmark. Final metrics will be reported directly from the frozen machine-readable TEST result record and are not inferred from the validation benchmark.

Method	Corrupted RMSE	Corrupted MAE	Pearson	Spearman
Traditional	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
MLP Gateway	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>

5.8 IPIP-120 Independent TEST

The independent TEST population contains 1,723,596 corrupted observations and 61,557 clean observations, for a total of 1,785,153 evaluated observations. The final global TEST metrics will be inserted directly from the frozen machine-readable TEST result record.

Table 8: IPIP-120 independent TEST head-to-head performance.

Method	Corrupted RMSE	Corrupted MAE	Pearson	Spearman
Traditional	–	–	–	–
MLP Gateway	–	–	–	–

The independent TEST benchmark was reserved from model selection, architecture screening, objective screening, and retraining. The final numerical values in this table will be populated only from the frozen machine-readable TEST result record.

5.9 IPIP-300 Validation

The IPIP-300 validation benchmark contains 338,444 corrupted observations in addition to 21,808 clean observations, for a total of 360,252 evaluated validation observations. Traditional scoring reproduces the canonical reference representation exactly on clean validation observations.

Table 9: IPIP-300 frozen validation head-to-head performance.

Method	Corrupted RMSE	Corrupted MAE	Pearson	Spearman
Traditional	0.240123	0.092047	0.842268	0.839497
MLP Gateway	0.188995	0.113684	0.899300	0.888280

The frozen benchmark reports an absolute corrupted-RMSE reduction of 0.051128 and a relative reduction of 21.293%. Pearson correspondence increases from 0.842268 to 0.899300, while Spearman correspondence increases from 0.839497 to 0.888280. Traditional scoring retains lower corrupted MAE.

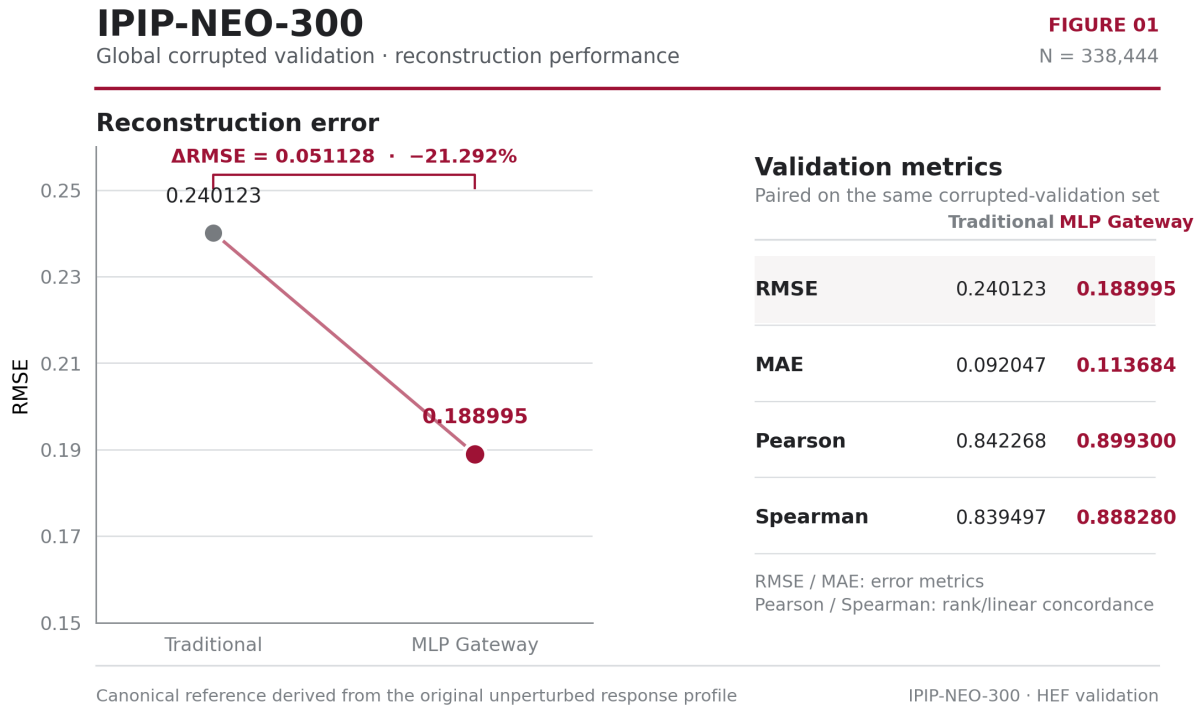


Figure 7: Global corrupted-validation performance for IPIP-NEO-300.

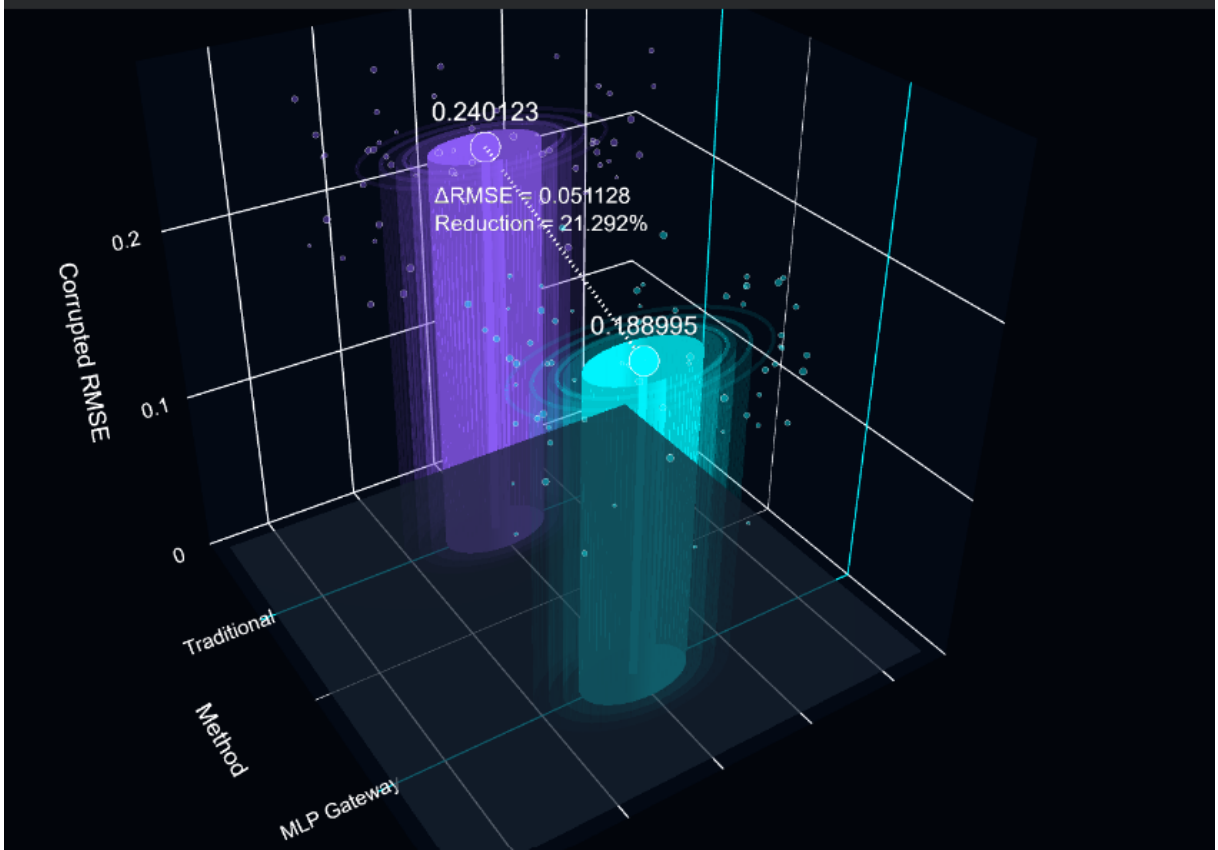


Figure 8: Three-dimensional visualization of the global corrupted-validation RMSE comparison for IPIP-NEO-300. The visualization presents the same aggregate reconstruction quantities evaluated against the canonical reference representation.

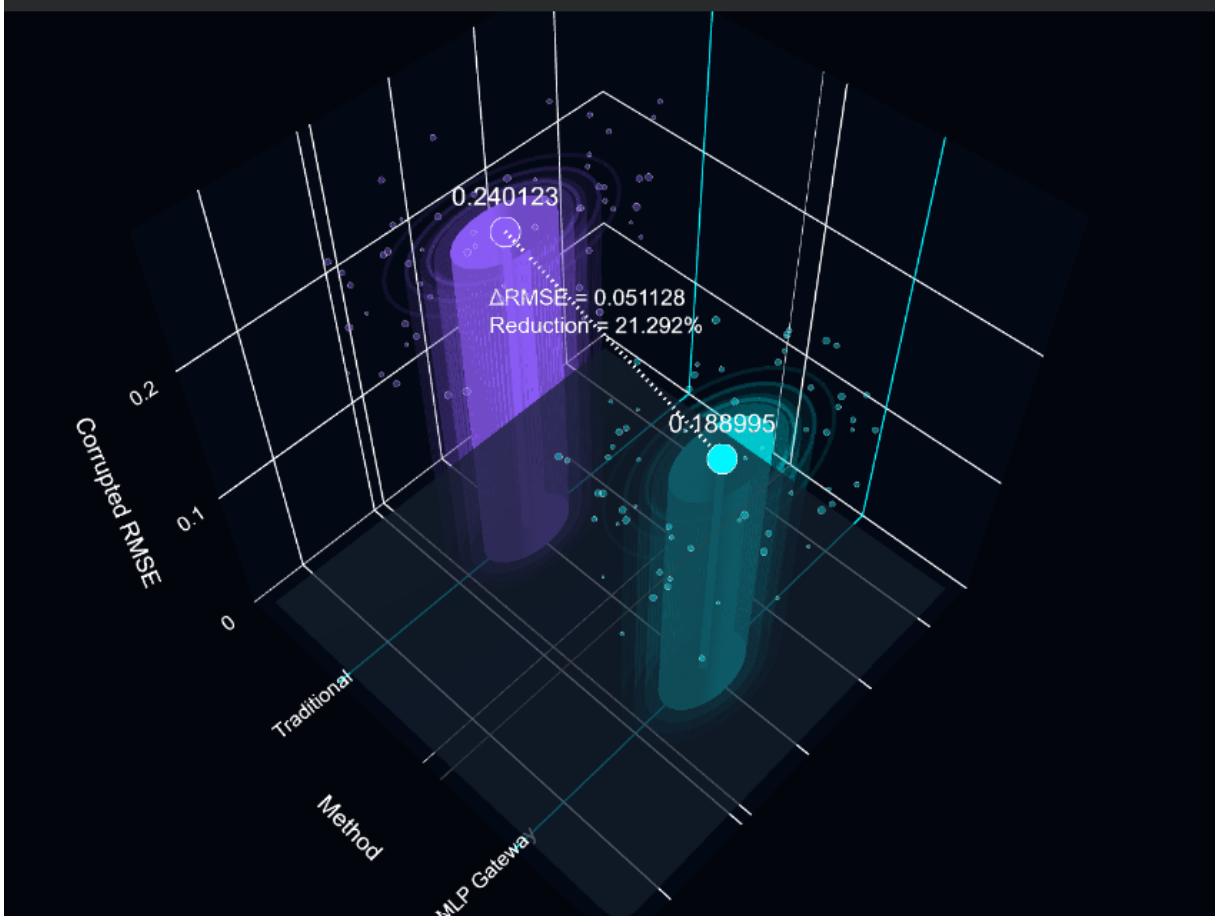


Figure 9: Complementary visualization of the global corrupted-validation comparison between Traditional scoring and the MLP Gateway for IPiP-NEO-300. The visualization represents the same frozen validation benchmark without introducing an additional evaluation condition.

At the row level, the MLP Gateway is closer to the canonical reference than Traditional scoring for 53.13% of corrupted validation rows, whereas Traditional scoring is closer for 46.87%; ties are absent.

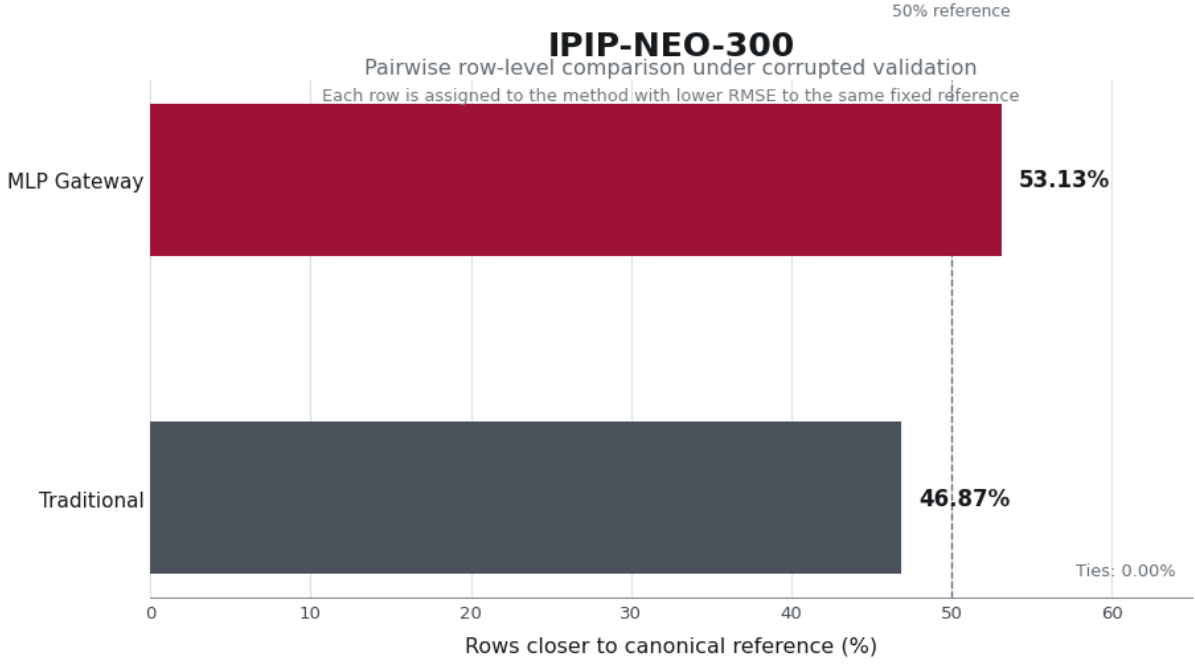


Figure 10: Paired row-level comparison of distance to the canonical reference under IPIP-NEO-300 corrupted validation.

5.10 Cross-Instrument Results

The two configurations show a consistent directional pattern under their respective validation protocols: lower corrupted-validation RMSE and higher Pearson and Spearman correspondence for the learned Gateway, accompanied by lower corrupted MAE for Traditional scoring.

Table 10: Cross-instrument validation summary.

Configuration	Traditional RMSE	Gateway RMSE	Relative RMSE reduction
IPIP-120	0.462274	0.347833	24.756%
IPIP-300	0.240123	0.188995	21.293%

These percentages are not pooled into a single effect size because the instruments, architectures, and validation populations differ.

The IPIP-120 configuration uses the compact Gateway architecture, whereas the IPIP-300 configuration uses the structured-output Gateway. Accordingly, the two experiments constitute complementary demonstrations of the reconstruction formulation rather than a single replicated architecture experiment.

5.11 Corruption-Family and Severity Analysis

Family-specific and severity-specific analyses provide evidence about the heterogeneity of the reconstruction effect across perturbation structure.

For IPIP-300, the MLP Gateway achieves lower RMSE in six of the seven corruption families. The only family in which Traditional scoring has lower RMSE is `ITEM_SWAP`. This exception is retained explicitly rather than being suppressed.

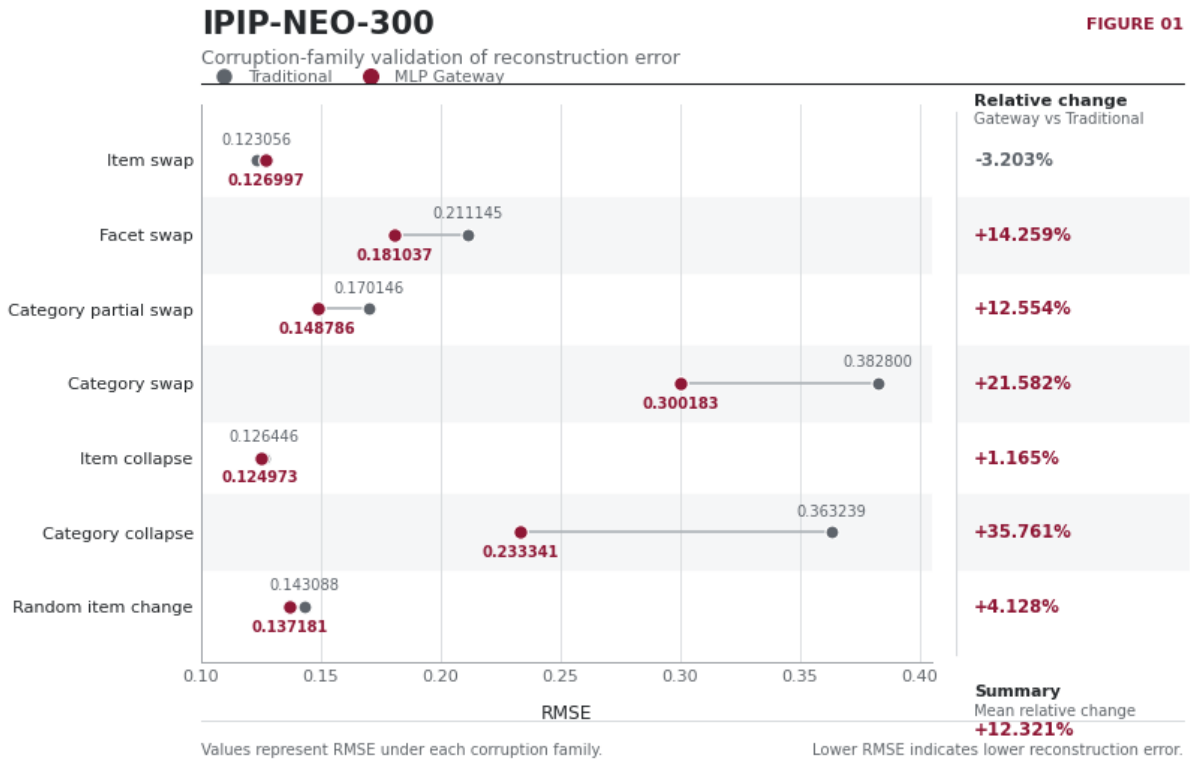


Figure 11: Corrupted RMSE across the seven IPIP-NEO-300 corruption families.

The relative RMSE reduction varies substantially across corruption families. The largest reported reduction occurs for `CATEGORY_COLLAPSE`, whereas `ITEM_SWAP` is the single family with a negative Gateway reduction relative to Traditional scoring.

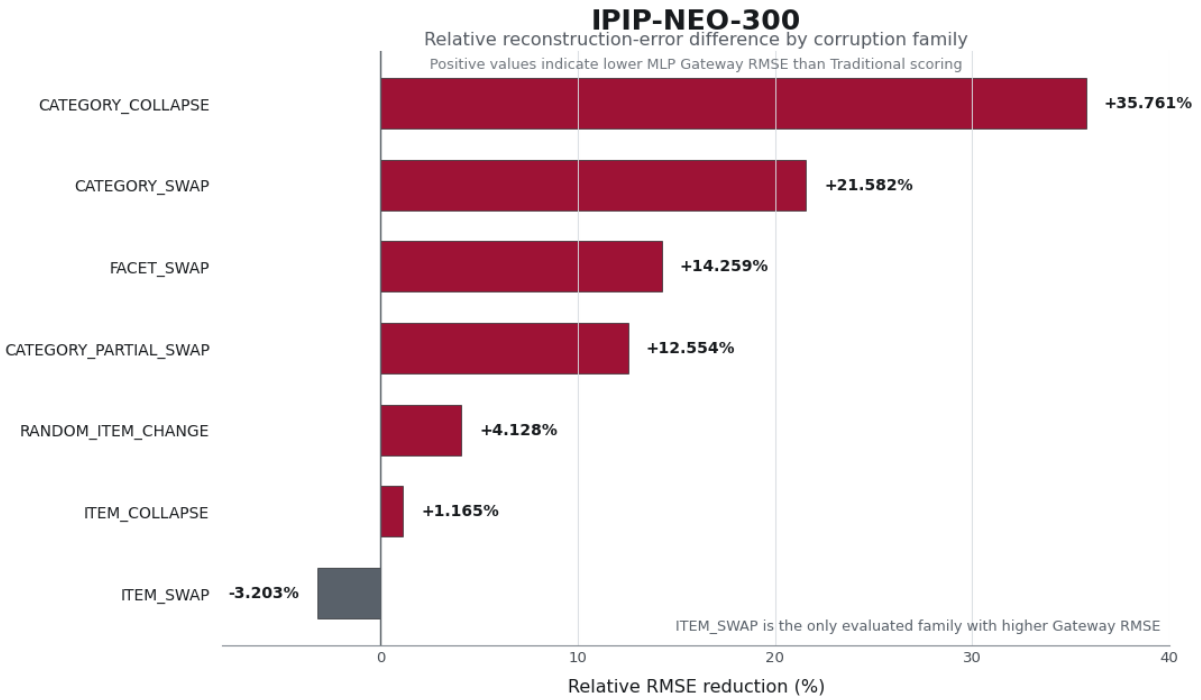


Figure 12: Relative corrupted-RMSE reduction of the MLP Gateway by IPIP-NEO-300 corruption family.

Across the thirteen evaluated severity values, the MLP Gateway achieves lower RMSE at twelve levels. The only exception is severity 12, at which Traditional scoring has slightly lower RMSE. This condition remains visible in the severity analysis.

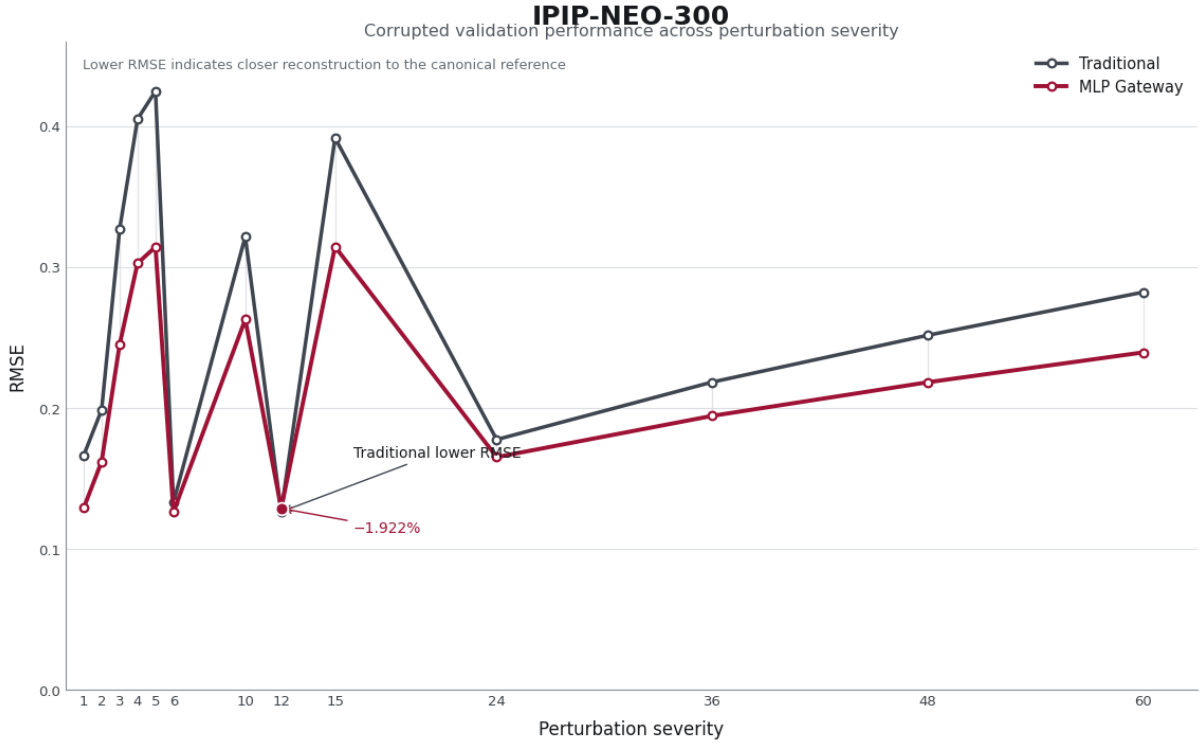


Figure 13: Corrupted RMSE across the evaluated IPIP-NEO-300 perturbation severity values.

The relative reduction is heterogeneous across severity. In particular, the severity-12 condition yields a small negative relative reduction, while the remaining evaluated severity values show positive relative reductions.

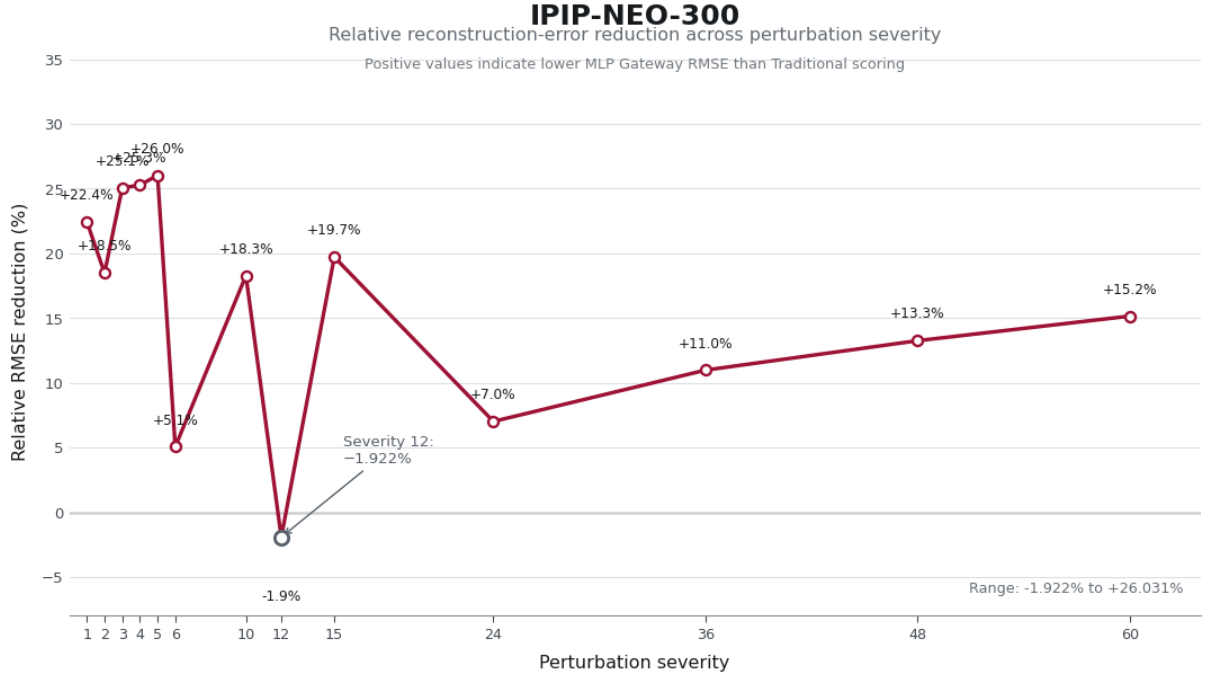


Figure 14: Relative corrupted-RMSE reduction of the MLP Gateway across IPIP-NEO-300 perturbation severity.

Taken together, the family- and severity-level analyses indicate that the observed robustness effect is broad but not uniform across every perturbation condition. The explicit exceptions are therefore part of the empirical result rather than artifacts to be removed from the presentation.

5.12 Facet-Level Analysis

The facet-level analysis evaluates whether the aggregate robustness pattern is concentrated in a subset of outputs or distributed across the 30-dimensional canonical psychometric representation.

For IPIP-300, the MLP Gateway achieves lower RMSE than Traditional scoring in all 30 facets. The magnitude of the reduction varies across facets, indicating that the aggregate effect is distributed across the output space but is not homogeneous in magnitude.

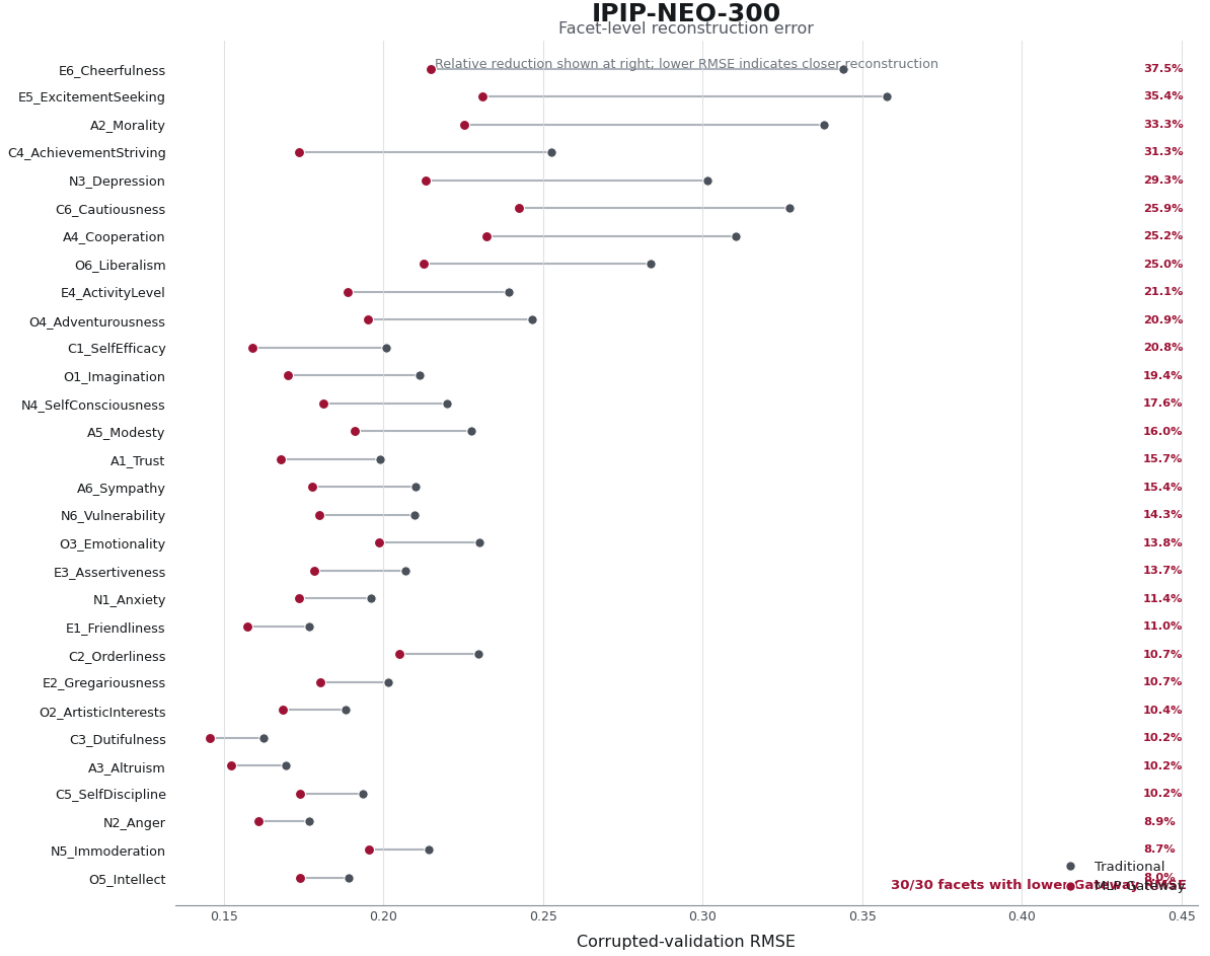


Figure 15: Facet-level reconstruction RMSE for the 30 IPIP-NEO-300 facets under corrupted validation.

The complete facet-level results are retained in the machine-readable benchmark record, with the plotted comparison providing the principal visual summary of the 30-dimensional reconstruction behavior.

6. Representation and Geometric Analysis

The IPIP-300 mechanistic laboratory evaluates whether structured input corruption is accompanied by measurable changes in the learned internal representation and whether these changes are empirically associated with output-space displacement and reconstruction error.

The analysis is organized as a sequence of distinct, non-equivalent measurements:

$$\text{input perturbation} \rightarrow \text{latent displacement} \rightarrow \text{geometric diagnostics} \rightarrow \text{output-space displacement} \rightarrow \text{reconstruction error} \quad (76)$$

This sequence is descriptive rather than causal. In particular, latent displacement, geometric deformation, output-space displacement, and reconstruction error are treated as distinct quantities and are not used interchangeably.

6.1 Latent Representation Geometry

The Gateway exposes hidden representations at dimensions 256, 128, and 64. Let

$$\mathbf{h}^{(256)}, \quad \mathbf{h}^{(128)}, \quad \mathbf{h}^{(64)} \quad (77)$$

denote the corresponding hidden representations for a given response vector. These representations are evaluated for matched clean and corrupted inputs.

The resulting latent spaces are treated as observational representations of the learned mapping. Their geometry is characterized using complementary descriptors including latent displacement, covariance structure, CKA, and local-neighborhood preservation.

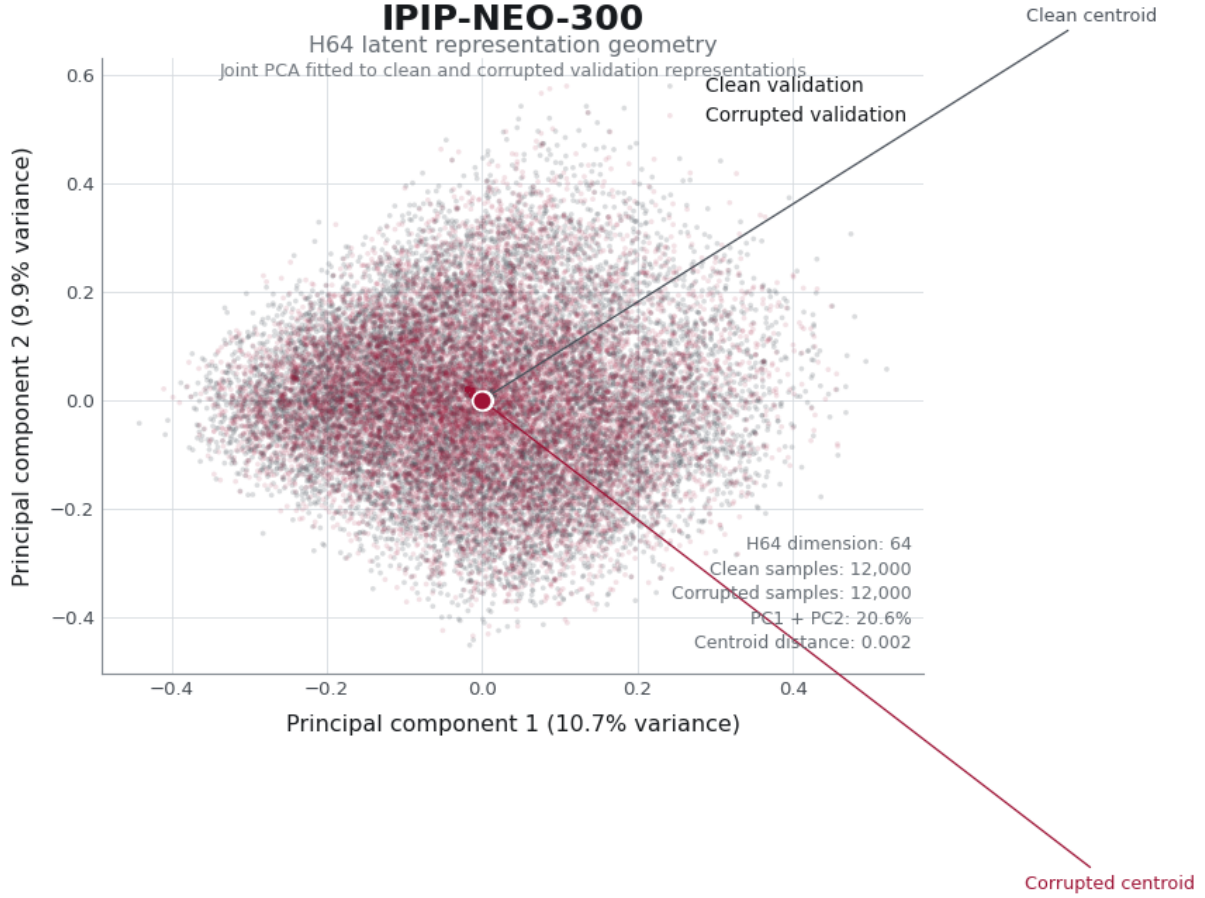


Figure 16: Latent representation geometry of the IPIP-NEO-300 Gateway across the analyzed hidden representations.

6.2 Latent Displacement

For a clean representation $\mathbf{h}^{(d)}$ and its matched corrupted counterpart $\tilde{\mathbf{h}}^{(d)}$, where $d \in \{256, 128, 64\}$, relative latent displacement is defined as

$$D_{\text{rel}}^{(d)} = \frac{\|\tilde{\mathbf{h}}^{(d)} - \mathbf{h}^{(d)}\|_2}{\|\mathbf{h}^{(d)}\|_2 + \epsilon}, \quad (78)$$

where ϵ is a numerical stabilizer.

The mechanistic analysis evaluates $D_{\text{rel}}^{(256)}$, $D_{\text{rel}}^{(128)}$, and $D_{\text{rel}}^{(64)}$ separately. Angular similarity is evaluated as a complementary geometric quantity using row-wise cosine similarity,

$$C^{(d)} = \frac{\mathbf{h}^{(d)} \cdot \tilde{\mathbf{h}}^{(d)}}{\|\mathbf{h}^{(d)}\|_2 \|\tilde{\mathbf{h}}^{(d)}\|_2}, \quad (79)$$

and is not interpreted as equivalent to Euclidean displacement.

Relative latent displacement therefore measures the magnitude of movement between matched clean and corrupted representations, whereas cosine similarity measures their angular alignment. Neither quantity is itself a reconstruction-error measure, since reconstruction error is defined separately with respect to the fixed canonical reference representation.

6.3 Spectral Structure

The covariance structure of each hidden representation is examined after centering the corresponding representation matrix. Let $\lambda_1, \dots, \lambda_r$ denote the non-negative eigenvalues of the resulting covariance matrix, ordered in descending magnitude, and define

$$p_i = \frac{\lambda_i}{\sum_j \lambda_j}. \quad (80)$$

Effective dimensionality is summarized using the entropy effective rank

$$R_{\text{eff}} = \exp \left(- \sum_{i:p_i > 0} p_i \log p_i \right), \quad (81)$$

and the participation ratio

$$R_{\text{PR}} = \frac{1}{\sum_i p_i^2}. \quad (82)$$

These quantities distinguish the nominal representation dimension from the effective dimensionality occupied by the observed representation geometry.

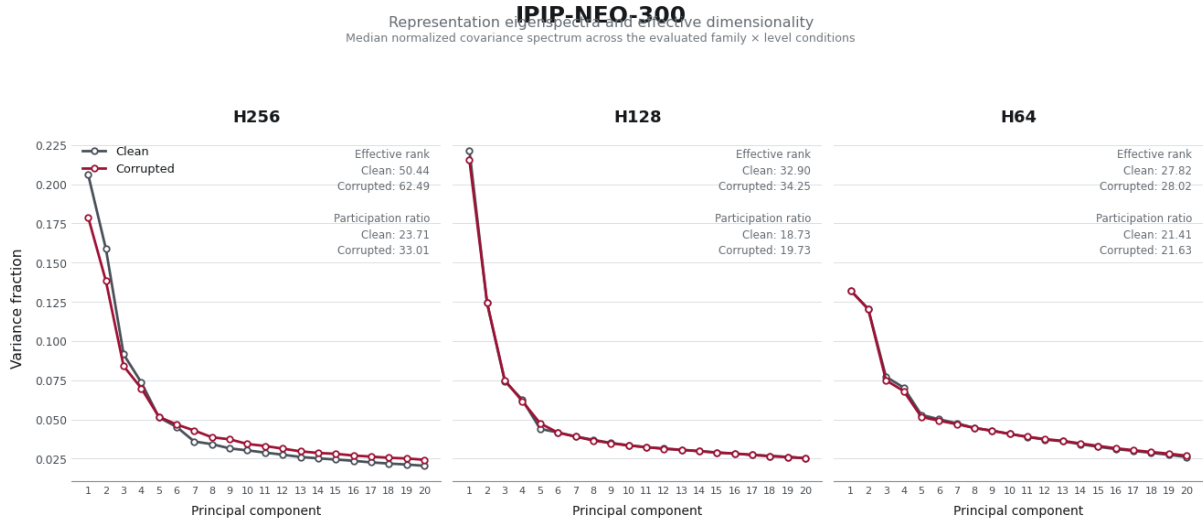


Figure 17: Covariance eigenspectra and effective representational dimensionality for the analyzed IPIP-NEO-300 hidden representations.

6.4 Spectral-Diffusion Geometry

The learned Gateway graph is further examined through a diffusion-based spectral representation of its computational structure. This analysis provides a multiscale geometric description complementary to the covariance spectrum and effective-dimensionality measures reported above.

Let P denote the diffusion operator constructed from the graph representation of the frozen Gateway, and let

$$P\psi_\ell = \lambda_\ell \psi_\ell \quad (83)$$

denote its eigenpairs. A diffusion-coordinate representation at diffusion time t can be expressed generically as

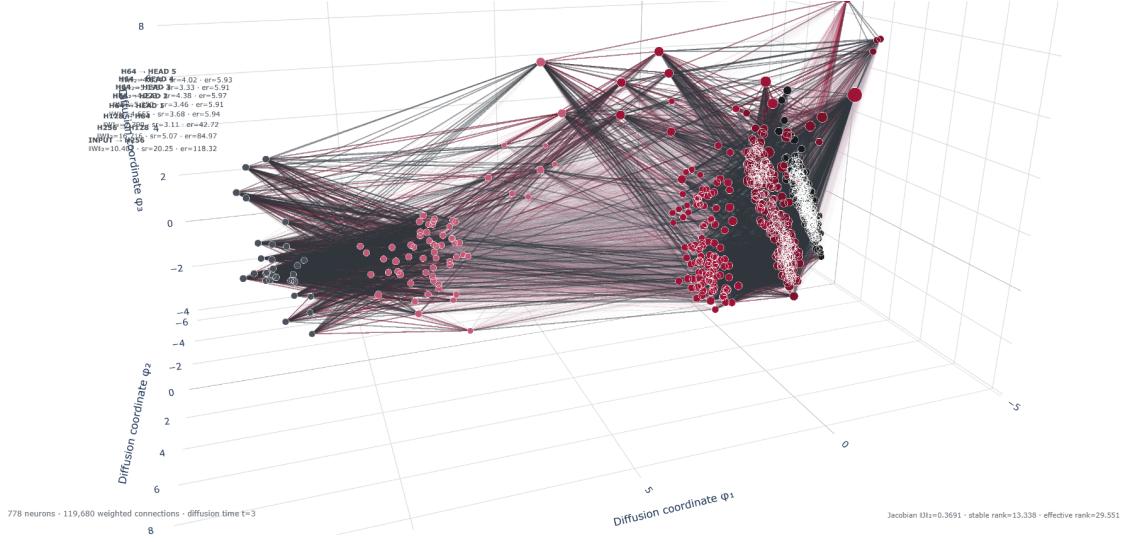
$$\Phi_t(v) = \left(\lambda_1^t \psi_1(v), \lambda_2^t \psi_2(v), \dots, \lambda_q^t \psi_q(v) \right), \quad (84)$$

where v denotes a graph element and the leading non-trivial spectral components are retained for visualization. Diffusion-map constructions use such eigenfunctions to obtain low-dimensional coordinates that preserve multiscale geometric information induced by the underlying diffusion process (**coifman2006diffusion**).

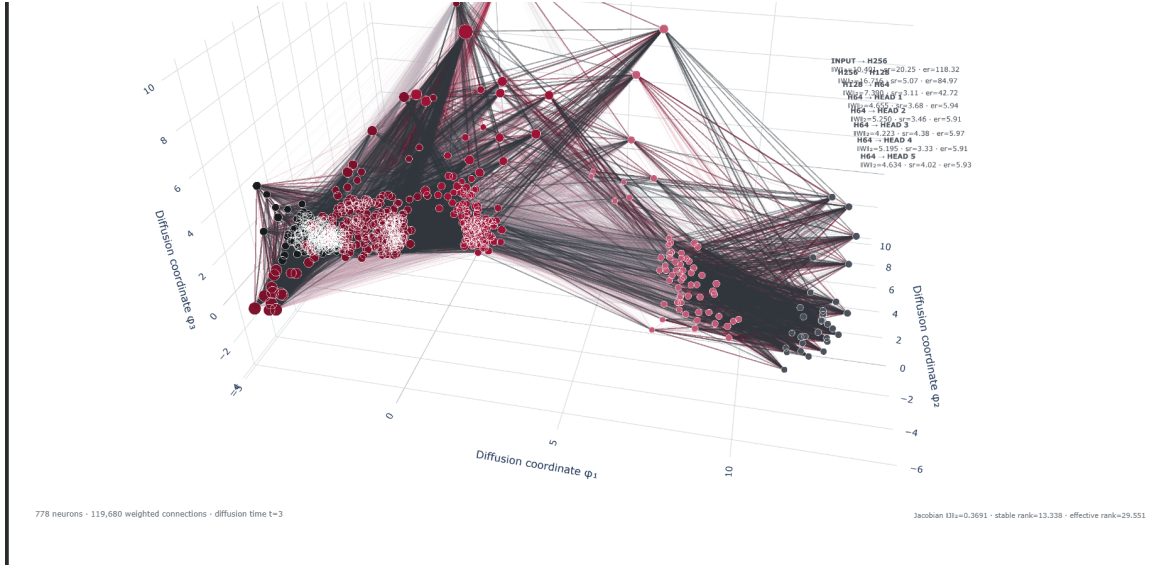
For the present visualization, the first three reported diffusion coordinates are denoted

$$\Phi(v) = (\phi_1(v), \phi_2(v), \phi_3(v)), \quad (85)$$

with the analysis evaluated at diffusion time $t = 3$.



(a) Primary three-dimensional diffusion-coordinate view.



(b) Complementary viewing angle of the same diffusion representation.

Figure 18: Spectral-diffusion representation of the frozen IPIP-NEO-300 MLP Gateway graph at diffusion time $t = 3$. The two panels show the same operator-derived geometric representation from complementary viewing angles. The visualization retains the full neuron-level graph structure while projecting its geometry into the leading diffusion coordinates (ϕ_1, ϕ_2, ϕ_3) .

The diffusion coordinates provide a geometric embedding of the graph that is distinct from the original Euclidean coordinates used to arrange the neurons for visualization. They are therefore interpreted as coordinates induced by the spectral properties of the diffusion operator rather than as physical spatial locations.

The analysis also records operator-level diagnostics associated with the learned mapping. Let $J(\mathbf{x})$ denote the Jacobian of the Gateway mapping with respect to its input. Its operator norm is

$$\|J(\mathbf{x})\|_2 = \sigma_{\max}(J(\mathbf{x})), \quad (86)$$

where $\sigma_{\max}$ denotes the largest singular value. This quantity summarizes the maximum local first-order amplification of perturbations in the linearized mapping.

For a matrix A , stable rank is defined as

$$\text{srank}(A) = \frac{\|A\|_F^2}{\|A\|_2^2}, \quad (87)$$

which provides a continuous measure of effective matrix dimensionality that is less sensitive to small singular values than the algebraic rank.

The entropy-based effective rank used in the representation analysis is

$$R_{\text{eff}} = \exp \left(- \sum_i p_i \log p_i \right), \quad (88)$$

where

$$p_i = \frac{\lambda_i}{\sum_j \lambda_j}. \quad (89)$$

Together, the diffusion coordinates, Jacobian norm, stable rank, and effective rank characterize different aspects of the learned computational geometry: multiscale graph organization, local first-order sensitivity, and effective dimensionality. These diagnostics are descriptive properties of the frozen numerical model and are not interpreted as evidence of an underlying psychological topology or as causal explanations of reconstruction performance.

6.5 Persistent Homology

The latent representation is additionally examined using persistent homology to characterize topological structure across a range of distance scales.

For a filtered point cloud, persistent homology tracks the appearance and disappearance of topological features across the filtration. The resulting persistence diagram represents each detected feature by its birth and death coordinates,

$$(b_i, d_i), \quad (90)$$

where b_i denotes the scale at which the feature appears and d_i denotes the scale at which it disappears.

The analysis distinguishes connected components, represented by the zeroth homology group H_0 , from one-dimensional topological structures represented by the first homology group H_1 . The lifetime of a finite persistence feature is

$$\ell_i = d_i - b_i, \quad (91)$$

which provides a scale-dependent measure of its persistence within the filtration.

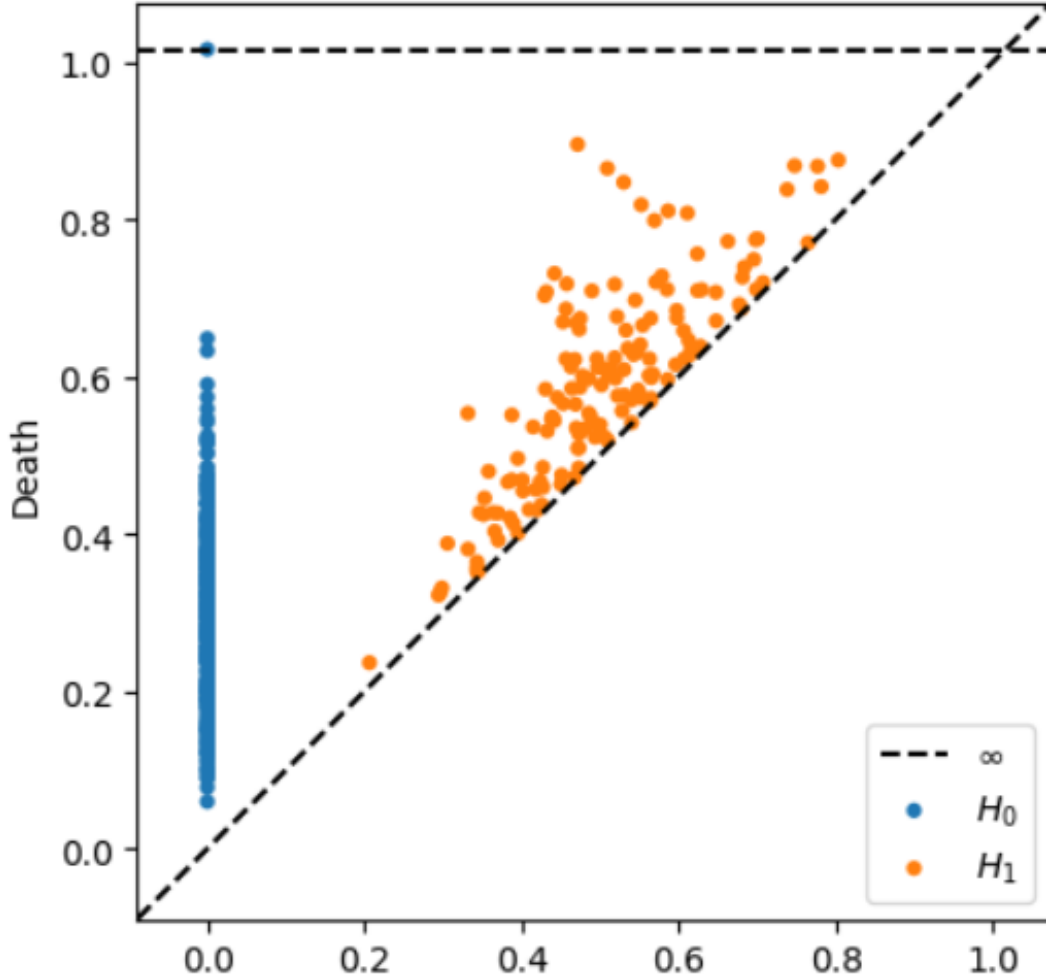


Figure 19: Persistence diagram of the analyzed latent representation. Features are represented in birth–death coordinates and separated by homological dimension into connected components (H_0) and one-dimensional structures (H_1). The diagonal corresponds to zero persistence, whereas points farther from the diagonal represent features with longer persistence across the filtration.

The persistence diagram provides a topological summary that is complementary to covariance spectra, local-neighborhood preservation, and representation alignment. It is not interpreted as evidence of a specific psychological topology; rather, it characterizes the topology of the numerical representation generated by the learned mapping under the evaluated experimental conditions.

6.6 Centered Kernel Alignment

Linear Centered Kernel Alignment (CKA) is used to compare the structure of matched clean and corrupted representations at H256, H128, and H64.

For centered representation matrices A and B , linear CKA is computed as

$$\text{CKA}_{\text{lin}}(A, B) = \frac{\|A^\top B\|_F^2}{\|A^\top A\|_F \|B^\top B\|_F}. \quad (92)$$

Thus,

$$\text{CKA}_{256}, \quad \text{CKA}_{128}, \quad \text{CKA}_{64} \quad (93)$$

measure similarity between the matched clean and corrupted representation structures at the three hidden stages. CKA is interpreted as a structural similarity measure and is not treated as a measure of reconstruction accuracy.

Because the comparison is performed on matched clean and corrupted representations, the resulting CKA values characterize the extent to which the learned representational structure is preserved under the specified perturbations. Values closer to one indicate greater structural similarity, whereas lower values indicate greater representational reorganization under the corruption conditions evaluated.

6.7 Neighborhood Preservation

Local geometric preservation is evaluated in the H64 representation space. For each representation, cosine similarity is used to identify nearest neighbors after excluding the observation itself from the candidate neighborhood.

Neighborhood overlap is evaluated at

$$k \in \{5, 10, 20\}. \quad (94)$$

For a clean neighborhood set $\mathcal{N}_k^{clean}(i)$ and its corrupted counterpart $\mathcal{N}_k^{corr}(i)$, the row-wise overlap is defined as

$$O_k(i) = \frac{|\mathcal{N}_k^{clean}(i) \cap \mathcal{N}_k^{corr}(i)|}{k}. \quad (95)$$

The resulting quantity measures the fraction of nearest neighbors retained between matched clean and corrupted representations. The analysis therefore quantifies preservation of local representation geometry under corruption rather than preservation of psychological identity or any other external construct.

6.8 Output Representation Deformation

Output-space displacement and reconstruction error are treated as separate quantities.

Let $\mathbf{Y}_{Gateway}^{clean}$ and $\mathbf{Y}_{Gateway}^{corr}$ denote the clean and corrupted Gateway outputs. Their relative output-space displacement is

$$D_{out}^{Gateway} = \frac{\|\mathbf{Y}_{Gateway}^{corr} - \mathbf{Y}_{Gateway}^{clean}\|_2}{\|\mathbf{Y}_{Gateway}^{clean}\|_2 + \epsilon}. \quad (96)$$

Analogously, Traditional scoring produces

$$D_{out}^{Traditional} = \frac{\|\mathbf{Y}_{Traditional}^{corr} - \mathbf{Y}_{Traditional}^{clean}\|_2}{\|\mathbf{Y}_{Traditional}^{clean}\|_2 + \epsilon}. \quad (97)$$

These quantities measure how far the output moves relative to its clean counterpart. They are distinct from reconstruction error, which is defined with respect to the fixed canonical reference representation.

For method m , let

$$\mathbf{E}^{(m)} = \mathbf{Y}_m^{corr} - \mathbf{Y}^{ref}. \quad (98)$$

The corresponding reconstruction error is therefore evaluated as a function of the fixed reference target rather than as a function of the corrupted output itself. This distinction separates output-space displacement from agreement with the canonical reference representation.

6.9 Item-to-Output Propagation

The item-to-output analysis summarizes how perturbations involving each input item are associated with error across the 30 output facets.

The term “propagation” is used here as a descriptive shorthand for an aggregated input-to-output error association. It does not denote a causal effect or a causal transmission mechanism. Let

$$M_{ij}^{item \rightarrow output, m} \quad (99)$$

denote the aggregated mean absolute reconstruction error observed at output facet j under corruption conditions involving input item i for method m .

The mechanistic laboratory retains separate matrices for the MLP Gateway and Traditional scoring, as well as their difference,

$$\Delta M_{ij}^{item \rightarrow output} = M_{ij}^{item \rightarrow output, Gateway} - M_{ij}^{item \rightarrow output, Traditional}. \quad (100)$$

These matrices provide a comparative description of the output-error structure associated with item-level perturbations. They are interpreted as error-localization summaries and not as evidence of causal transmission from individual questionnaire items to specific output facets.

6.10 Facet-to-Output Propagation

A complementary analysis aggregates perturbations at the input-facet level. The resulting matrix is 30-by-30, with input facets indexing rows and output facets indexing columns.

Let

$$M_{ij}^{facet \rightarrow output, m} \quad (101)$$

denote the aggregated mean absolute output error for output facet j associated with perturbation conditions affecting input facet i for method m .

As above, the analysis retains separate Gateway and Traditional matrices and their difference,

$$\Delta M_{ij}^{facet \rightarrow output} = M_{ij}^{facet \rightarrow output, Gateway} - M_{ij}^{facet \rightarrow output, Traditional}. \quad (102)$$

The resulting structure is interpreted as a comparative error-localization diagnostic describing how perturbations associated with different input facets correspond to error across the output representation. It is not interpreted as evidence of causal propagation between personality facets.

6.11 Output Error Correlation

Output error correlation is defined from the residual vectors relative to the fixed canonical reference.

For method m ,

$$\mathbf{E}^{(m)} = \hat{\mathbf{Y}}^{(m)} - \mathbf{Y}^{ref}, \quad (103)$$

and the 30-by-30 error-correlation matrix is

$$R^{(m)} = \text{Corr}(\mathbf{E}^{(m)}). \quad (104)$$

The laboratory evaluates these structures separately for the MLP Gateway and Traditional scoring. Correlated output errors therefore describe the organization of residual error across the 30-dimensional representation. They are not themselves measures of psychological association, latent construct correlation, or psychometric validity.

6.12 Regression and Partial Correlation Analysis

The mechanistic laboratory further evaluates whether reconstruction error is statistically associated with corruption severity, latent displacement, output-space displacement, and corruption-family indicators.

At the row level, a generic pooled specification can be written as

$$RMSE_i = \beta_0 + \beta_1 S_i + \beta_2 D_i^{(256)} + \beta_3 D_i^{(128)} + \beta_4 D_i^{(64)} + \beta_5 D_i^{out} + \boldsymbol{\gamma}^T \mathbf{F}_i + \varepsilon_i, \quad (105)$$

where S_i denotes corruption severity, the $D_i^{(d)}$ terms denote latent relative displacement at the corresponding hidden dimension, D_i^{out} denotes output-space displacement, and $\mathbf{F}_i$ represents corruption-family indicators.

The mechanistic laboratory records the original validation participant target identifier as the clustering variable and includes within-subject structure in the pooled analysis. Accordingly, repeated corrupted observations from the same original participant are not treated as independent observations for cluster-robust inference.

Partial correlations are reported only with respect to explicitly specified conditioning variables. In general form,

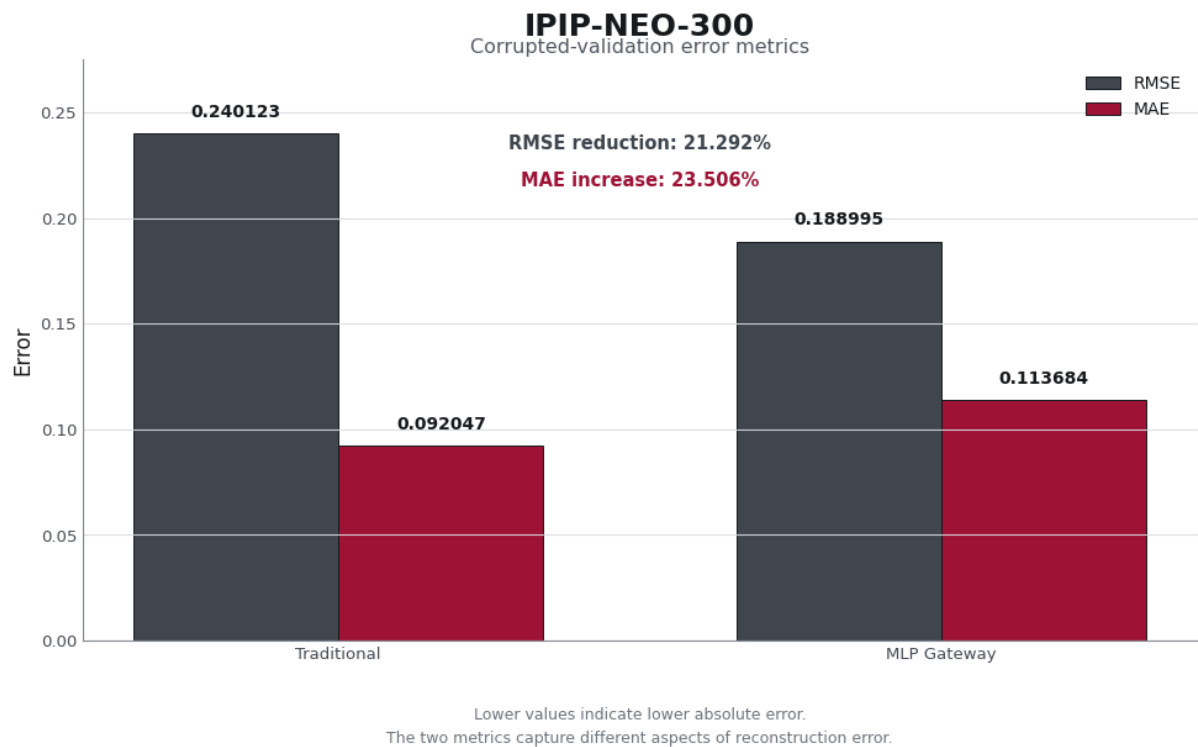
$$\rho(X, Y \mid \mathbf{Z}) \quad (106)$$

denotes the association between X and Y after conditioning on the variables in $\mathbf{Z}$. The final reported conditioning sets are specified together with the corresponding machine-readable coefficient and partial-correlation records.

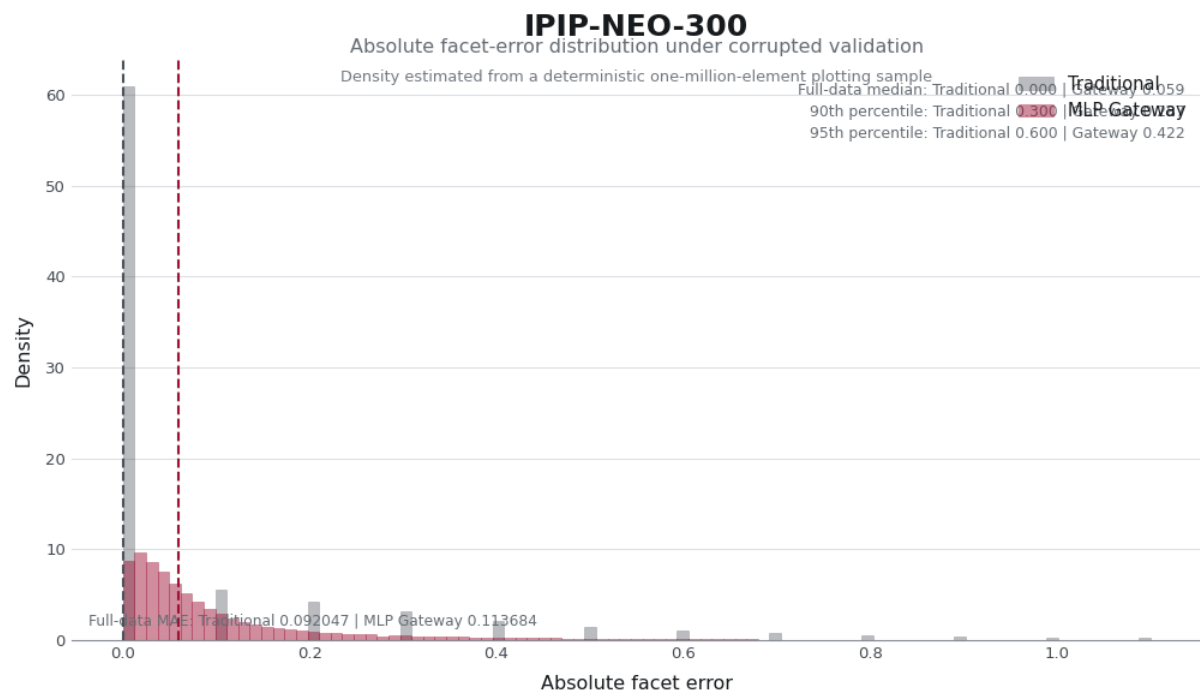
Throughout these analyses, latent displacement, geometric similarity, output-space displacement, and reconstruction error remain distinct quantities. The resulting regression and partial-correlation analyses are therefore interpreted as associational evidence concerning the geometry of degradation and the organization of reconstruction error, rather than as evidence that latent displacement causes output error.

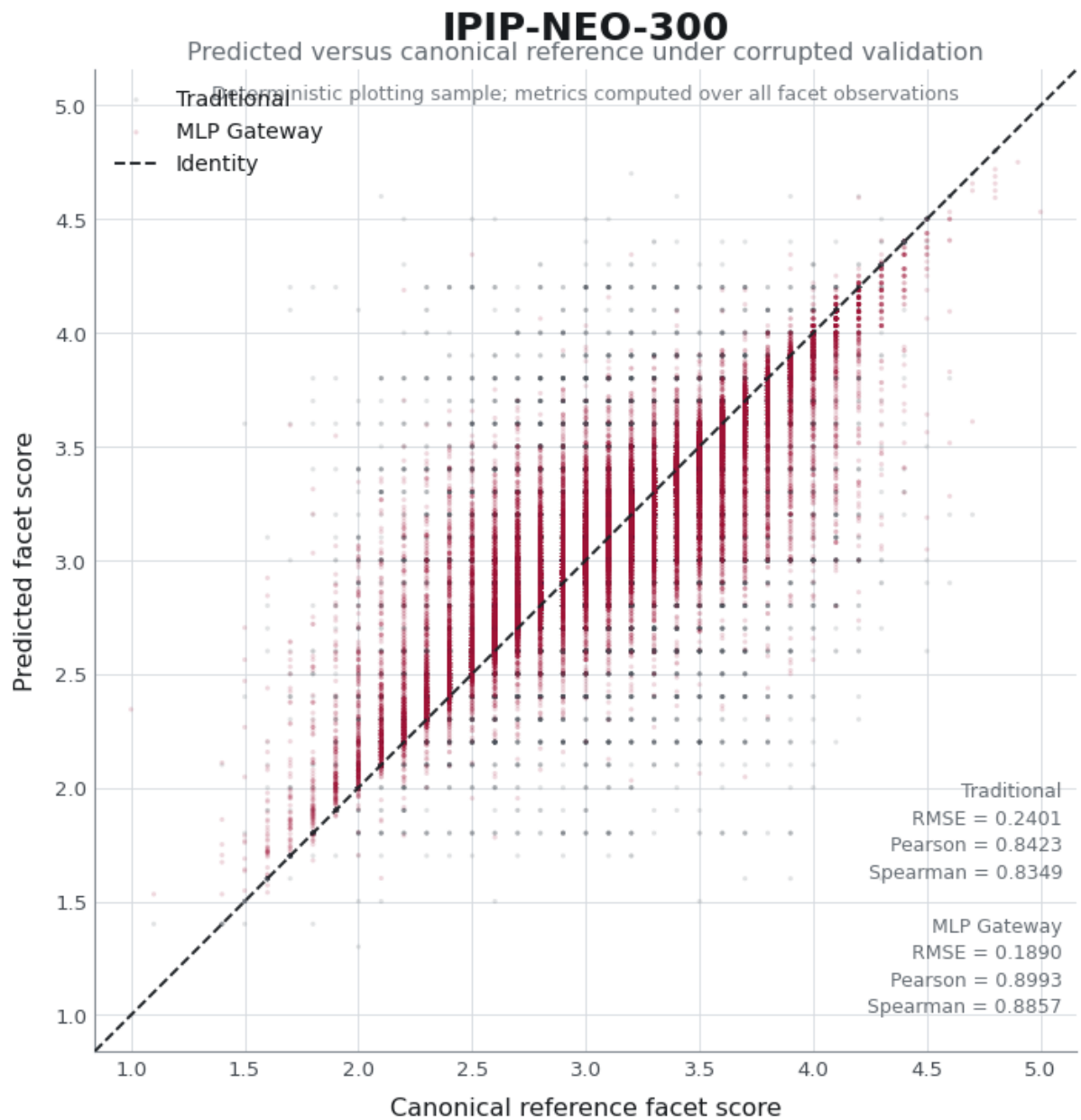
7. Error Analysis

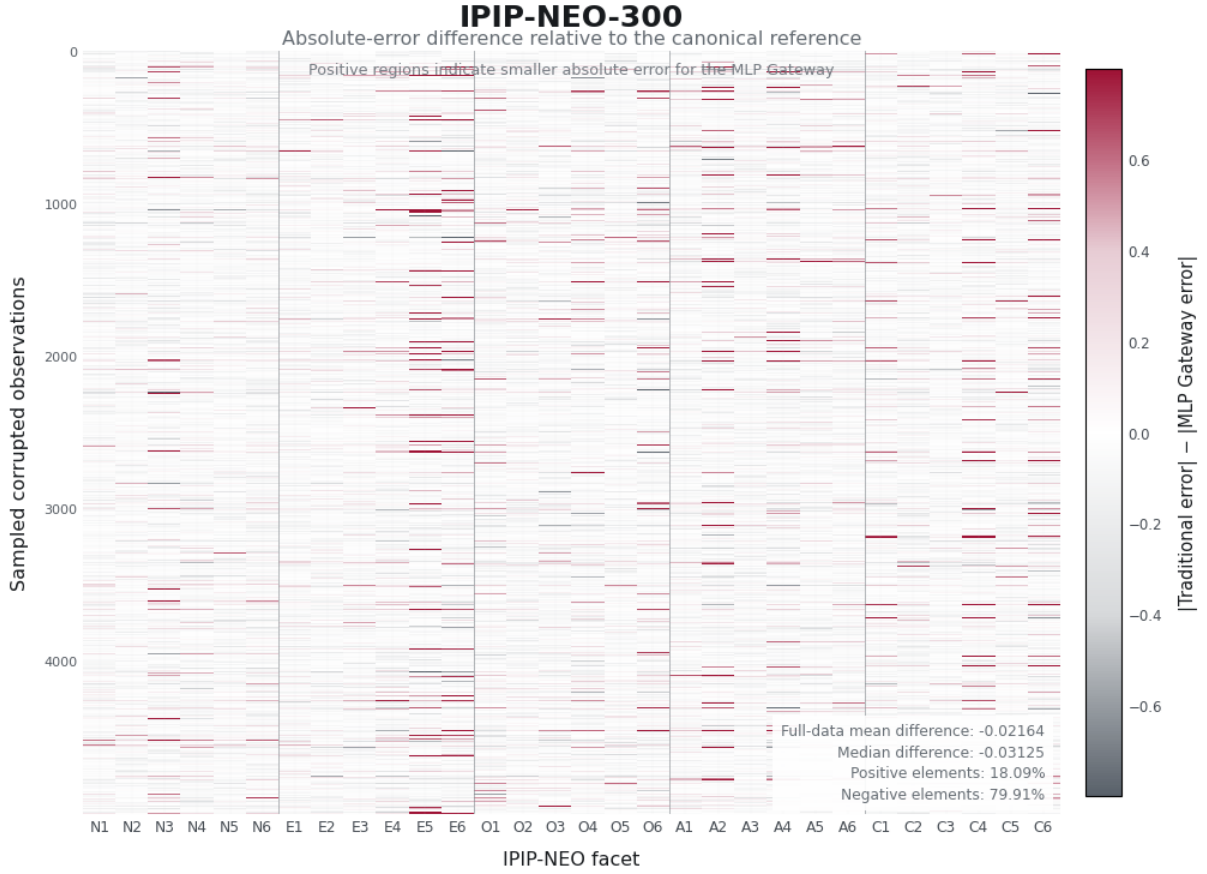
7.1 RMSE–MAE Trade-Off



The principal robustness signal is metric-specific. Lower RMSE occurs together with higher corrupted MAE in both instrument configurations. This distinction indicates that the error distribution should be inspected rather than summarized through a single metric.







7.2 Error Tails

Upper-tail behavior is evaluated from the empirical distribution of row-level reconstruction errors. This analysis complements global averages such as RMSE and MAE by quantifying the magnitude and frequency of unusually large reconstruction errors.

Let $r_i^{(m)}$ denote the row-level reconstruction error for observation i under method m , where the row-level metric is computed against the fixed canonical reference representation. The empirical cumulative distribution function is

$$\hat{F}^{(m)}(t) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(r_i^{(m)} \leq t). \quad (107)$$

The empirical upper-tail quantile at probability $p \in (0, 1)$ is defined as

$$Q_p^{(m)} = \inf \left\{ t : \hat{F}^{(m)}(t) \geq p \right\}. \quad (108)$$

Upper-tail diagnostics are evaluated at prespecified probabilities collected in the set

$$\mathcal{P}_{\text{tail}} = \{p_1, p_2, \dots, p_K\}, \quad (109)$$

with the exact values fixed by the computational protocol. Because different statistical software packages implement sample quantiles using different definitions and interpolation conventions, the quantile definition and implementation are fixed for reproducibility and retained as part of the analysis specification (**hyndman1996quantiles**).

For a predefined high-error threshold τ , the corresponding tail proportion is

$$\pi_{\tau}^{(m)} = \frac{1}{N} \sum_{i=1}^N \mathbb{I} \left(r_i^{(m)} > \tau \right). \quad (110)$$

When a threshold region is specified, the analysis therefore reports both the threshold exceedance rate and the corresponding error magnitude. These quantities describe the concentration of large errors and are not interpreted as separate estimates of clinical or psychometric validity.

7.3 Output Range

The Gateway is evaluated without output clipping. Predictions outside the nominal scoring domain are therefore retained rather than projected back into the admissible interval, allowing the behavior of the learned mapping near and beyond the reference output domain to be measured directly.

For the 30-dimensional facet representation, the nominal coordinate range is

$$\mathcal{Y} = [1, 5]^{30}, \quad (111)$$

corresponding to facet scores obtained from the five-point response scale used by the evaluated IPIP-NEO configurations (**goldberg1999ipip**; **johnson2014ipipneo120**).

For predicted output $\hat{\mathbf{Y}}_i = (\hat{y}_{i1}, \dots, \hat{y}_{i30})$, an out-of-range indicator is defined for each output coordinate as

$$I_{ij}^{out} = \mathbb{I}(\hat{y}_{ij} < 1 \vee \hat{y}_{ij} > 5). \quad (112)$$

The overall out-of-range rate is then

$$R_{out}^{(m)} = \frac{1}{30N} \sum_{i=1}^N \sum_{j=1}^{30} I_{ij}^{out}. \quad (113)$$

Out-of-range magnitude is additionally quantified by

$$M_{out}^{(m)} = \frac{1}{30N} \sum_{i=1}^N \sum_{j=1}^{30} [\max(0, 1 - \hat{y}_{ij}) + \max(0, \hat{y}_{ij} - 5)]. \quad (114)$$

Thus, $R_{out}^{(m)}$ measures the frequency of violations of the nominal scoring domain, whereas $M_{out}^{(m)}$ measures their mean magnitude. Because no clipping is applied, these quantities characterize the unconstrained output behavior of the learned mapping rather than the behavior of a post-processed prediction.

7.4 Failure Localization

Failure localization is formulated as an error-concentration analysis over the perturbation families, severity levels, affected input items, affected input facets, and output facets represented in the experimental protocol. The purpose is to determine where reconstruction error is concentrated rather than to infer a causal source of the error. Such disaggregated evaluation is consistent with broader model-testing practice in which aggregate performance alone may fail to reveal important model-specific failure modes (**ribeiro2020checklist**).

For observation i and output facet j , define the absolute reconstruction error for method m as

$$e_{ij}^{(m)} = \left| \hat{y}_{ij}^{(m)} - y_{ij}^{ref} \right|. \quad (115)$$

For any analysis group G , let $\mathcal{I}_G$ denote the set of observations associated with that group. The conditional mean absolute error is

$$MAE_G^{(m)} = \frac{1}{30|\mathcal{I}_G|} \sum_{i \in \mathcal{I}_G} \sum_{j=1}^{30} e_{ij}^{(m)}. \quad (116)$$

The corresponding conditional root-mean-square error is

$$RMSE_G^{(m)} = \sqrt{\frac{1}{30|\mathcal{I}_G|} \sum_{i \in \mathcal{I}_G} \sum_{j=1}^{30} \left(e_{ij}^{(m)} \right)^2}. \quad (117)$$

These group-specific quantities are evaluated separately for corruption families and severity levels and, where the perturbation metadata identify the affected coordinates, for input items and input facets.

For output-facet localization, the error associated with facet j is

$$MAE_j^{(m)} = \frac{1}{N} \sum_{i=1}^N e_{ij}^{(m)}, \quad (118)$$

with the corresponding output-facet RMSE given by

$$RMSE_j^{(m)} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(e_{ij}^{(m)} \right)^2}. \quad (119)$$

To distinguish error magnitude from error frequency, an error-mass statistic is also defined for a specified group G as

$$S_G^{(m)} = \frac{\sum_{i,j} \mathbb{I}_G(i,j) \left(e_{ij}^{(m)} \right)^2}{\sum_{i,j} \left(e_{ij}^{(m)} \right)^2}, \quad (120)$$

where $\mathbb{I}_G(i,j)$ indicates whether the observation–output pair belongs to the corresponding diagnostic group.

The joint use of MAE_G , $RMSE_G$, and S_G separates three distinct properties: average error magnitude, sensitivity to large errors, and the fraction of total squared-error mass associated with a specified group. Because groups such as corruption family, severity, item, input facet, and output facet can differ in frequency and may overlap conceptually, no single quantity is interpreted as a causal attribution.

For method comparison, the same localization statistics are computed for the MLP Gateway and Traditional scoring relative to the same immutable canonical reference representation. The resulting analysis identifies the experimental conditions and output dimensions in which reconstruction error is concentrated while preserving the distinction between descriptive error localization and causal explanation.

8. Discussion

The empirical results support a specific methodological observation: under the controlled perturbation protocols evaluated here, a learned nonlinear mapping can exploit information from the complete questionnaire response vector to reduce relatively large reconstruction errors relative to direct deterministic scoring.

One interpretation is that the learned Gateway can exploit statistical redundancy extending across items and facets. Traditional scoring is constrained to the designated items for each facet, whereas the Gateway receives the complete response vector. The observed family-specific variation indicates, however, that the learned reconstruction is not uniformly insensitive to every perturbation structure.

The metric trade-off is central to the interpretation. Lower RMSE together with higher MAE is consistent with a redistribution of errors in which relatively large deviations are reduced while smaller deviations may remain or increase. The effect should therefore not be summarized as a universal reduction in every notion of error.

The cross-instrument results are complementary rather than a direct architectural replication. IPIP-120 uses the compact Gateway, whereas IPIP-300 uses the structured-output Gateway. The shared directional pattern across the two instrument-specific validation programs is therefore consistent with the generality of the reconstruction formulation under the evaluated conditions, but it does not isolate which architectural properties account for the observed behavior or establish architectural invariance across instruments.

The geometric analysis adds a second layer of evidence by examining whether corrupted observations are accompanied by measurable changes in internal representations and whether those changes are statistically associated with output-space displacement and reconstruction error. These analyses provide additional characterization of the learned representation but do not establish a causal mechanism linking latent deformation to reconstruction error.

The study also has a deliberately limited health interpretation. No patient cohort, clinical endpoint, diagnosis, treatment decision, or prospective clinical workflow was evaluated. Consequently, the current evidence concerns computational reconstruction of a canonical reference psychometric representation rather than clinical decision support.

9. Limitations

1. The perturbation protocol is controlled and semi-synthetic. It uses observed response patterns from real incomplete participants as donor material for selected transformations, but it does not establish robustness to every real-world missingness, careless-response, measurement-error, or adversarial mechanism.
2. The canonical reference representation is operationally defined by established deterministic scoring of the original, unperturbed response vector. It should not be interpreted as an independently observed latent psychological truth or as a clinical ground truth.
3. The IPIP-120 and IPIP-300 configurations use different Gateway architectures and should not be treated as a single repeated architecture experiment.
4. Validation model selection and independent testing are distinct stages. The IPIP-120 TEST is reported separately from validation, whereas the IPIP-300 study reports the frozen validation benchmark and its associated mechanistic analyses.

5. The Gateway does not currently provide calibrated uncertainty, normative percentile estimates, or validated clinical thresholds.
6. Output values are not clipped to the nominal $[1, 5]$ interval in the reported production inference.
7. The present study evaluates reconstruction agreement with the canonical reference representation rather than clinical validity, diagnostic performance, or prospective decision utility.
8. Population, language, demographic, and external-site generalization have not been established by these experiments alone.

10. Intended Use and Ethical Scope

The present work is intended as a methodological study of robust reconstruction in questionnaire-derived representations. The experiments do not justify autonomous diagnosis, clinical triage, treatment recommendation, or other medical decision-making.

Any future study involving clinical populations should independently establish population validity, measurement properties, uncertainty estimation, privacy protections, governance, and prospective utility before clinical use is considered.

The open-science objective of the project is to facilitate independent scrutiny, reproduction, extension, and validation rather than to bypass professional psychometric or clinical standards.

11. Reproducibility and Open Science

The study is designed around a frozen model-and-analysis contract. The public research package will contain source code, configuration files, model metadata, reproducibility documentation, machine-readable results, figure-generation code, and cryptographic hashes for frozen artifacts.

11.1 Code Availability

The reproducibility repository will contain the final IPIP-120 and IPIP-300 evaluation code together with analysis utilities and environment specifications.

11.2 Model Availability

The released production model files will be accompanied by architecture metadata and SHA-256 digests so that users can verify byte-level artifact identity.

11.3 Dataset Provenance

Original instrument materials and derived perturbation datasets will be documented separately. The paper will distinguish source instrument data, derived response representations, semi-synthetic perturbation data, donor material used to construct selected perturbations, and model outputs.

11.4 Versioning and Integrity Verification

The public preprint will identify a versioned release. Subsequent revisions will retain version history rather than silently replacing the initial record.

12. Conclusion

We formulated robust psychometric reconstruction under structured response perturbation as a finite-dimensional nonlinear function approximation problem and evaluated instrument-specific multilayer perceptron Gateway configurations on IPIP-NEO-120 and IPIP-NEO-300.

Across the frozen validation benchmarks, the learned Gateway exhibits lower corrupted-validation RMSE and higher Pearson and Spearman correspondence with the immutable canonical reference representation, while Traditional scoring retains lower corrupted MAE. The IPIP-120 program additionally includes an independent held-out TEST benchmark, and the IPIP-300 program provides a detailed representation and geometric analysis of the frozen validation model.

These results support a methodological robustness finding within the specified semi-synthetic perturbation framework. They do not establish clinical validity, diagnostic utility, or universal superiority across error criteria, instruments, populations, or untested perturbation mechanisms. The primary scientific contribution of the framework is therefore the formulation, controlled evaluation, and reproducible analysis of nonlinear reconstruction under structured observation-space perturbations.

Data Availability

The manuscript uses IPIP-derived instrument materials together with derived response representations and internally generated semi-synthetic perturbation data. Exact data-access, redistribution, and licensing terms will be documented in the final release according to the provenance and licensing conditions of the underlying resources.

Code Availability

Source code for the final reproducibility release will be made publicly available in the associated GitHub repository.

Model Availability

Frozen model checkpoints and integrity manifests will be released alongside the reproducibility package, subject to the final artifact distribution policy.

Author Contributions

Oscar Alejandro Laos Medina: conceptualization, methodology, software, dataset engineering, machine learning, validation, statistical analysis, visualization, and manuscript preparation.

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Competing Interests

The author declares no competing interests in the current manuscript version.

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A. Extended Model Specification

The final supplement will provide the complete instrument mappings, reverse-key audit, full training schedule, complete family- and severity-level tables, full facet-level results, geometric-analysis outputs, propagation matrices, regression tables, and machine-readable artifact references.

B. Artifact Registry

Table 11: Planned public research artifacts.

Artifact	Purpose
IPIP-120 Gateway checkpoint	Frozen compact reconstruction model
IPIP-300 Gateway checkpoint	Frozen structured-output reconstruction model
Model manifest	Architecture, objective, version, parameter count, provenance
SHA-256 registry	Integrity verification
Reproduction code	Independent rerun of preprocessing, inference, and evaluation
Figure-generation code	Reproduction of publication figures
Machine-readable tables	Exact global, family, severity, facet, and geometry results
Supplementary materials	Detailed methods and extended analyses